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Mitochondrial protein FgDML1 regulates DON toxin biosynthesis and cyazofamid sensitivity in *Fusarium graminearum* by affecting mitochondrial homeostasis

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eLife Assessment

This **important** study tackles an interesting aspect of fungal physiology: how a mitochondria-associated gene influences production of the secondary metabolite DON and fungicide sensitivity. The authors have improved the manuscript and the supporting evidence is **convincing**, although some uncertainties remain around descriptions of the methods.

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Abstract

Fusarium graminearum is a global pathogen responsible for Fusarium head blight (FHB) in wheat, causing substantial yield losses and producing the mycotoxin deoxynivalenol (DON), which poses a threat to both human and animal health. *Drosophila melanogaster* Misato-Like protein (DML1) plays a critical role in regulating mitochondrial function, yet its function in filamentous fungi remains unexplored. In this study, we characterized FgDML1 in *F. graminearum*. FgDML1 interacts with the mitochondrial fission and fusion protein FgDnm1 to maintain mitochondrial stability, thereby positively regulating acetyl-CoA levels and ATP synthesis, which influences toxosome formation and ultimately affects DON toxin biosynthesis. Additionally, FgDML1 is involved in the regulation of toxin biosynthetic enzyme expression. In the Δ FgDML1 mutant, Complex III enzyme activity decreased, overexpression of complex III assembly factors FgQCR2, FgQCR8, and FgQCR9 may induce conformational changes in the Qi-site protein, specifically altering the sensitivity of *F. graminearum* to respiratory inhibitor cyazofamid not Qo-site inhibitor pyraclostrobin and other fungicides. Furthermore, the loss of FgDML1 leads to defects in nutrient utilization, as well as in asexual and sexual reproduction, and pathogenicity. In conclusion, this study identifies a novel regulatory role for FgDML1 in DON toxin biosynthesis and cyazofamid sensitivity in *F. graminearum*. Our study provides a theoretical framework for understanding DON biosynthesis regulation in *F. graminearum* and identifies potential molecular targets for FHB control.

Introduction

Fusarium head blight (FHB), caused by *Fusarium graminearum*, is a devastating fungal disease that affects cereal crops, leading to significant yield losses in wheat-growing regions worldwide (Dean et al., 2012 [DOI](#)). This disease is particularly concerning due to the production of various mycotoxins, including trichothecenes and zearalenone, during fungal infection (Hai et al., 2023 [DOI](#)). Among these, deoxynivalenol (DON), a type of trichothecene, is the most commonly detected mycotoxin (Merhej et al., 2011 [DOI](#)). The DON toxin poses a serious threat to both human and animal

health, significantly compromising immune and reproductive systems (Knutsen et al., 2017). Chronic exposure to DON can lead to anorexia, neurological disorders, and immune dysregulation, while acute high-dose exposure may result in vomiting, diarrhea, neurological disturbances, and even miscarriage or stillbirth (Sumarah, 2022; Zhang et al., 2024). Studies have shown that increased DON levels are positively correlated with the pathogenicity rate of *F. graminearum* (Bönnighausen et al., 2019; Ilgen et al., 2009; Mudge et al., 2006). As FHB continues to spread and its severity increases, DON contamination is emerging as a critical global concern (Duan et al., 2020; Liu et al., 2016). This highlights the need for effective strategies to manage and mitigate the impact of both the disease and the associated mycotoxin contamination.

The absence of resistant cultivars necessitates reliance on chemical methods as the primary approach for controlling FHB (Liu et al., 2019). However, the inappropriate application of fungicides has led to the emergence of resistance, particularly against carbendazim, tebuconazole, and pyraclostrobin (Andrade et al., 2022; Yi et al., 2023; Zhou et al., 2024). Notably, carbendazim has been demonstrated to stimulate the biosynthesis of DON toxin, especially in fungicide-resistant populations of *F. graminearum* (Li et al., 2019; Zhou et al., 2020). Similarly, pyraclostrobin significantly promotes DON biosynthesis, with no selectivity between resistant and sensitive strains. Although tebuconazole has not been directly implicated in inducing DON production, it is classified as a fungicide with a moderate risk of resistance development (Duan et al., 2020). Field observations have revealed diverse resistance mechanisms to tebuconazole, complicating management efforts (Liu et al., 2011; Zhao et al., 2021). The potential evolution of resistance to tebuconazole is particularly concerning, as it could undermine its efficacy and exacerbate the challenge of controlling DON contamination. There is an urgent need for a fungicide that has not yet been used in *F. graminearum* to address the DON toxin contamination caused by fungicide resistance. Cyazofamid is a mitochondrial respiratory chain complex III inhibitor with demonstrated efficacy in controlling fungal diseases. It specifically binds to the Qi site of complex III, blocking the reduction of ubiquinone and inhibiting ATP synthesis, thereby exerting its fungicidal effect (Li et al., 2014; Mitani et al., 2001). However, it remains unclear whether other proteins within the intricate regulatory network of the respiratory chain play a role in modulating fungal sensitivity to cyazofamid. Further investigations are needed to elucidate the potential involvement of these regulatory components.

The synthesis of DON toxin is primarily regulated by genes involved in trichothecene biosynthesis, known as *FgTRI*s. Many of these *FgTRI* genes have been well characterized (Merhej et al., 2011). The initial step in DON synthesis is catalyzed by the trichothecene synthase *FgTri5*, which facilitates the cyclization of farnesyl pyrophosphate (FPP) to form trichodiene (TDN) (Kimura et al., 2007). However, before this catalytic process can occur, the organism must first obtain sufficient ATP from the mitochondria. ATP is critical not only for the basic cellular functions but also for driving several key reactions in the isoprenoid biosynthesis pathway. Specifically, ATP is required for the enzymatic conversion of precursors, leading to the production of FPP, a vital intermediate in the trichothecene biosynthesis pathway. The availability of ATP therefore directly impacts the efficient synthesis of FPP, which, in turn, regulates the downstream production of DON toxin. Without an adequate supply of ATP, these biosynthetic pathways cannot proceed efficiently, thus hindering the synthesis of both the precursor molecules and the final toxin product (Alexander et al., 2009). Pathogen-intrinsic ATP homeostasis is recognized as a critical, rate-limiting determinant for toxin biosynthesis. Previous studies indicate that dual-target inhibition of ATP synthase (AtpA) and adenine deaminase (Ade) by a specific small-molecule probe effectively depletes intracellular ATP, consequently suppressing the synthesis of key virulence factors TcdA and TcdB transcriptionally and translationally (Marreddy et al., 2024). The systemic toxicity of Anthrax Edema Toxin (ET) is primarily attributed to its catalytic activity, which depletes the host cell's ATP reservoir, thereby triggering a bioenergetic collapse that culminates in cell lysis and death (Liu et al., 2025). The biosynthesis of DON entails a reorganization of the endoplasmic reticulum into a specialized compartment termed the "toxosome" (Tang et al., 2018). The assembly of the toxosome coincides with the aggregation of key biosynthetic enzymes, which in turn enhances the efficiency of DON production. Concurrently, this compartmentalization serves

as a self-defense mechanism, protecting the fungus from the autotoxicity of TRI pathway intermediates (Boenisch et al., 2017 [↗](#)). The proteins TRI1, TRI4, TRI14, and Hmr1 are confirmed constituents of this structure (Kistler and Broz, 2015 [↗](#); Menke et al., 2013 [↗](#)).

Drosophila melanogaster Misato-Like protein (DML1) encodes a Misato-like protein in *Saccharomyces cerevisiae*, an essential factor involved in mitochondrial DNA (mtDNA) inheritance. In addition to its role in mtDNA maintenance, DML is implicated in mitochondrial compartmentalization and chromosome segregation. The protein shares structural features with members of the GTPase family, suggesting potential involvement in GTP-dependent processes (Gurvitz et al., 2002 [↗](#)). In *Homo sapiens*, the DML protein is localized to the mitochondria, where silencing of DML results in a transition from a networked mitochondrial morphology to a fragmented form. This indicates that DML plays a critical role in mitochondrial distribution, morphological fusion, and overall mitochondrial dynamics (Kimura and Okano, 2007 [↗](#)). In prokaryotes, the homologous protein FtsZ, which shares functional similarities with DML, has been identified as a target for antimicrobial agents (Huang et al., 2007 [↗](#); Vollmer, 2006 [↗](#)). However, despite these insights from studies in the aforementioned organisms, the exact role of DML in filamentous fungi remains undetermined.

This study systematically characterized the functional roles of DML protein in *F. graminearum* (FgDML1), including mycelial growth, asexual development, and sexual reproduction. More importantly, we found that FgDML1 reduces DON toxin synthesis by decreasing the enzyme activity of complex III, reducing ATP synthesis, and lowering acetyl-CoA accumulation. Furthermore, the deletion of FgDML1 results in compensatory upregulation of complex III assembly factors, which may alter the conformation of the Qi site protein and subsequently modulates sensitivity to cyazofamid.

Results

Identification, deletion, and complementation of FgDML1

The DML1 protein from *S. cerevisiae* was used as a query for a BLAST search against the *Fusarium* genome database, resulting in the identification of the putative DML1 gene FgDML1 (FGSG_05390) in *F. graminearum*. The FgDML1 gene is 1455 bp in length, contains three introns, and encodes a 484-amino acid protein with 23.88% homology to its counterpart in *S. cerevisiae*. Phylogenetic analysis was performed using DML1 sequences from other species obtained via NCBI, and a phylogenetic tree was constructed (Fig. 1A [↗](#)). Structural analysis revealed the presence of the Misat_Tub_SegII and Tubulin_3 domains. Furthermore, subcellular localization studies confirmed that FgDML1 localizes to mitochondria, as demonstrated by colocalization with a mitochondria-specific dye MitoTracker Red CMXRos (Fig. 1B [↗](#)).

To investigate the function of FgDML1, Δ FgDML1 deletion mutant was constructed and validated via PCR and Southern blot analysis (Fig. S2 [↗](#)), confirming the mutant as a single-copy knockout. To ensure that the observed phenotypes were specifically attributable to the deletion of FgDML1, an in situ complemented mutant, Δ FgDML1-C, was generated to restore FgDML1.

FgDML1 regulates vegetative growth

To assess the role of FgDML1 in vegetative growth, PH-1, Δ FgDML1, and Δ FgDML1-C were cultured on various media, including PDA, CM, MM, and V8, to measure growth rates. The Δ FgDML1 strain displayed a distinct growth phenotype characterized by retardation in radial growth and the formation of more compact, denser hyphal networks on all tested media compared to the PH-1 and Δ FgDML1-C strains. (Fig. 2A [↗](#)). Moreover, Δ FgDML1 displayed pronounced hyphal curvature, which was not observed in PH-1 or Δ FgDML1-C (Fig. 2B [↗](#)). These findings demonstrate that FgDML1 is a positive regulator of virulence in *F. graminearum*.

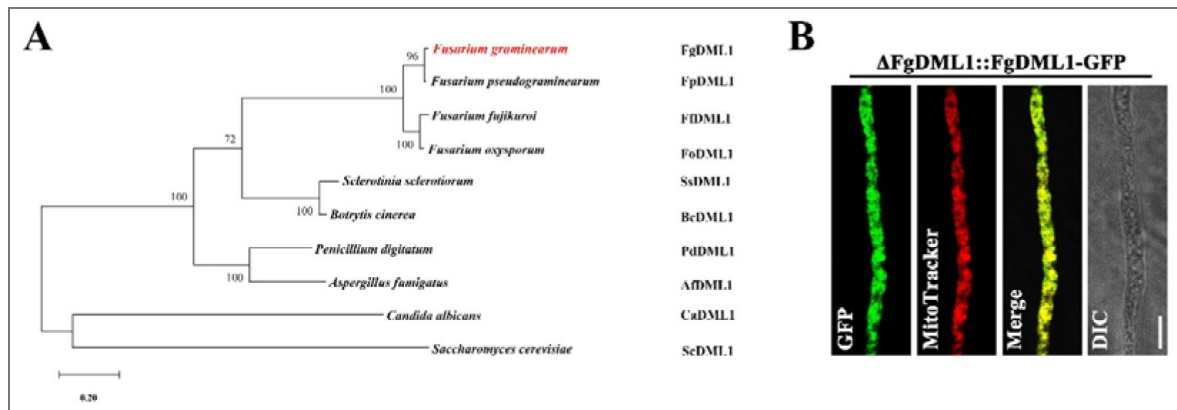


Figure 1. Identification and localization of FgDML1 in *Fusarium graminearum*.

(A) The phylogenetic tree was constructed based on the amino acid sequences of DML1 homologs from 10 species using the maximum likelihood method in Mega X. Bootstrap values from 1000 replications were displayed on the branches. (B) FgDML1 was localized to the mitochondria.

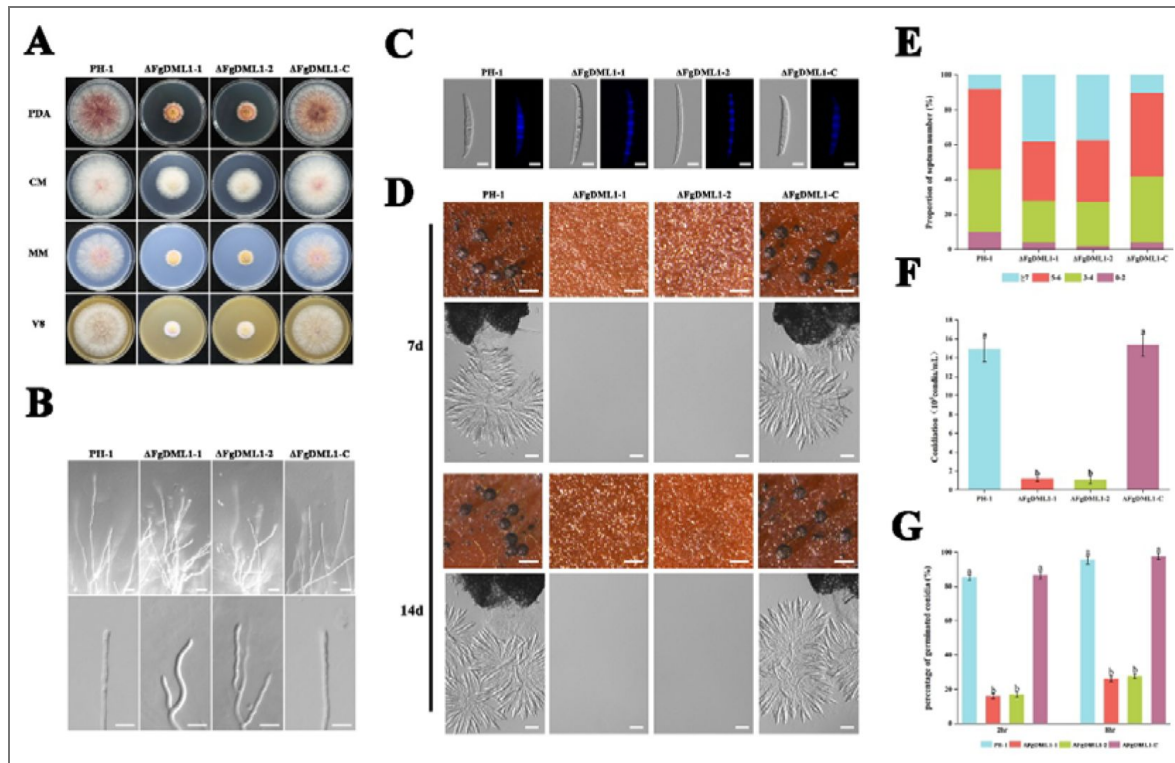


Figure 2. Effects of $\Delta FgDML1$ on asexual and sexual development.

(A) Mycelial growth rate. PH-1, $\Delta FgDML1$, $\Delta FgDML1-C$ were cultured for 3 days on PDA, CM, MM, and V8 solid media at 25°C. (B) Mycelial morphology. The mycelial morphology of PH-1, $\Delta FgDML1$, and $\Delta FgDML1-C$ was examined on WA plates. Scale bar = 72 μm . (C) Conidial length. Conidia of $\Delta FgDML1$ were significantly longer in length compared to PH-1 and $\Delta FgDML1-C$. Conidia were germinated in MBL medium, stained with Calcofluor White (CFW), and observed using an Olympus IX-71 microscope. (D) Sexual reproduction. PH-1 and $\Delta FgDML1-C$ produced perithecia and asci containing ascospores after 7 or 14 days of incubation on CA medium, while the $\Delta FgDML1$ failed to produce these structures under the same conditions. Perithecia and ascospores were observed using a Nikon SMZ25 fluorescent stereo microscope and an Olympus IX-71 inverted fluorescence microscope. Top image: perithecium, scale bar = 1 mm; bottom image: ascospores, scale bar = 50 μm . (E) Proportion of conidial septa. The proportion of conidial septa was assessed in the tested strains. (F) Conidial production. Conidia were harvested after 4 days of incubation in MBL medium, and their concentration was quantified using a hemocytometer. (G) Germination rate. Conidial germination was observed by incubating 300 conidia for 2 or 8 hours on WA plates. Germination rates were compared, with different letters indicating significant differences based on ANOVA followed by the LSD test ($p=0.05$).

FgDML1 plays a critical role in both asexual and sexual reproduction

To examine the role of FgDML1 in asexual reproduction, a series of experiments were conducted on conidia. Morphological analysis revealed that conidia from the Δ FgDML1 mutant were thinner and longer than those from the PH-1 and Δ FgDML1-C (Fig. 2C [↗](#)). Furthermore, statistical analysis of septum number indicated a significant increase in conidia with ≥ 7 septa in the Δ FgDML1 compared to the controls (Fig. 2E [↗](#)). Conidial production was significantly reduced in Δ FgDML1, with fewer spores produced compared to PH-1 and Δ FgDML1-C (Fig. 2F [↗](#)). The germination rate of Δ FgDML1 conidia was also significantly lower than that of the controls (Fig. 2G [↗](#)). Regarding sexual reproduction, Δ FgDML1 failed to produce perithecia or ascospores on both the 7th and 14th days of incubation (Fig. 2D [↗](#)).

FgDML1 is essential for virulence

To investigate the role of FgDML1 in virulence, pathogenicity assays were performed on wheat coleoptiles and wheat leaves. The results revealed that Δ FgDML1 caused only restricted infection at the inoculation site on wheat coleoptiles and failed to induce systemic infection, in contrast to PH-1 and Δ FgDML1-C, which exhibited extensive colonization (Fig. 3A, B [↗](#)). Similarly, Δ FgDML1 displayed a near-complete loss of virulence on wheat leaves (Fig. 3C, D [↗](#)). These findings demonstrate that FgDML1 is a critical positive regulator of virulence in *F. graminearum*.

FgDML1 is essential for the biosynthesis of the DON toxin

To evaluate whether the deletion of FgDML1 influences DON production, DON levels were measured under toxin-inducing conditions (28°C, 145 × g, 7 days incubation). The results revealed that DON production in Δ FgDML1 was reduced by nearly 80% compared to PH-1 and significantly lower than that of both PH-1 and Δ FgDML1-C (Fig. 4A [↗](#)). To further elucidate the regulatory relationship between FgDML1 and DON biosynthesis, the transcriptional levels of *FgTri5* and *FgTri6* were analyzed and found to be significantly downregulated in Δ FgDML1 (Fig. 4B [↗](#)). Moreover, examination of toxisomes in the mutant demonstrated that their formation was severely impaired, with almost no detectable toxisomes in Δ FgDML1 (Fig. 4C [↗](#)). Western blot analysis of the DON toxin marker protein FgTri1 further confirmed its markedly reduced levels in Δ FgDML1 compared to PH-1 (Fig. 4D [↗](#)). These findings suggest that FgDML1 is essential for regulating DON production in *F. graminearum*.

The deletion of FgDML1 results in reduced synthesis of both ATP and acetyl-CoA

The results showed that ATP levels in Δ FgDML1 were significantly lower compared to PH-1 and Δ FgDML1-C (Fig. 5D [↗](#)). Previous studies have established a positive correlation between ATP levels and acetyl-CoA. Consistent with this, acetyl-CoA levels in Δ FgDML1 were also significantly reduced compared to the controls (Fig. 5C [↗](#)). In conclusion, the deletion of FgDML1 leads to a reduction in both the key energy molecule ATP and the DON precursor acetyl-CoA.

FgDML1 regulates mitochondrial morphology by interacting with FgDnm1

To assess whether the loss of FgDML1 induces mitochondrial alterations, mitochondria-specific dyes were utilized, revealing fragmented mitochondria in the Δ FgDML1 mutant, whereas PH-1 and Δ FgDML1-C displayed a typical network-like mitochondrial distribution (Fig. 5A [↗](#)). Further analysis via Transmission Electron Microscope (TEM) identified damaged mitochondrial structures in the Δ FgDML1 mutant, in contrast to the well-preserved mitochondrial morphology observed in the control strains (Fig. 5B [↗](#)). Given that FgDnm1 is known to be a critical protein for

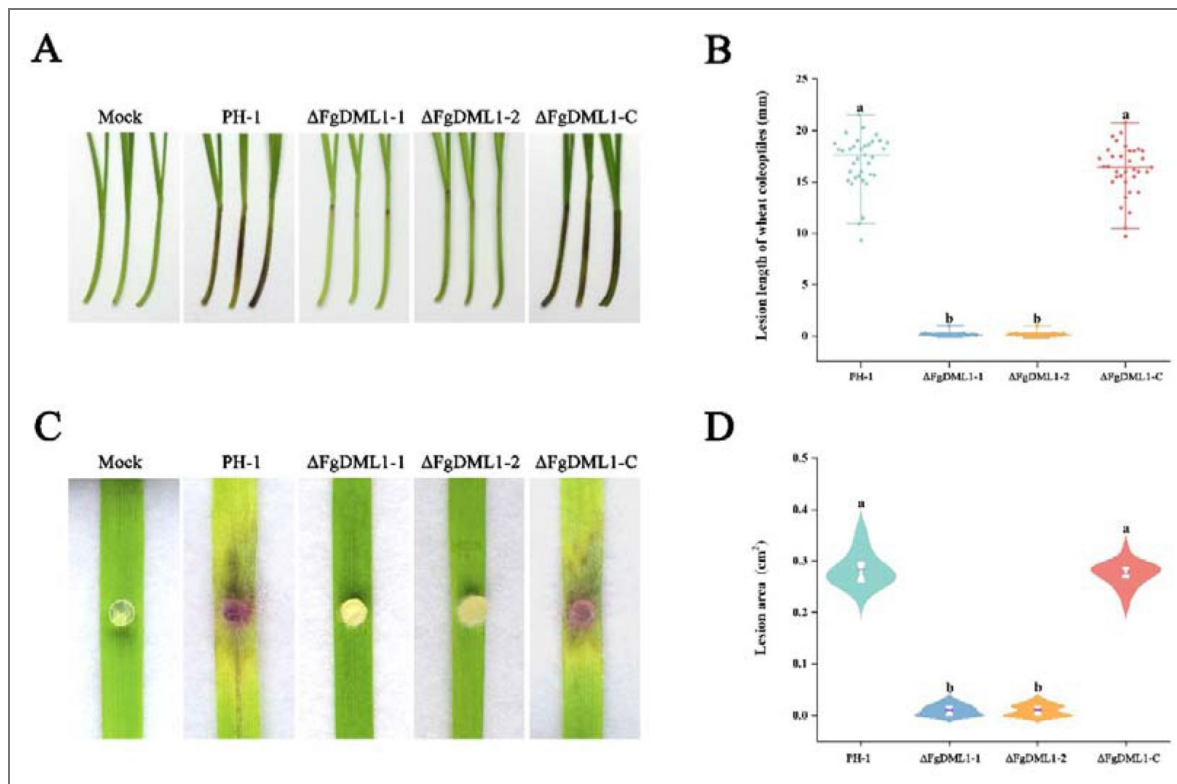


Figure 3. FgDML1 is required for both the virulence of *Fusarium graminearum*.

(A) Lesion lengths on wheat coleoptiles caused by different strains. These images were taken at 14 dpi. (B) Dot plot showing the lesion lengths on wheat coleoptiles at 14 dpi tested with different strains. (C) Disease symptoms on wheat leaves inoculated with PH-1, Δ FgDML, and Δ FgDML-C. Images were taken at 5 dpi. (D) Disease area on wheat leaves inoculated with different strains at 5 dpi. Bars with the same letter indicate no significant difference according to a significant difference (LSD) test at $p = 0.05$.

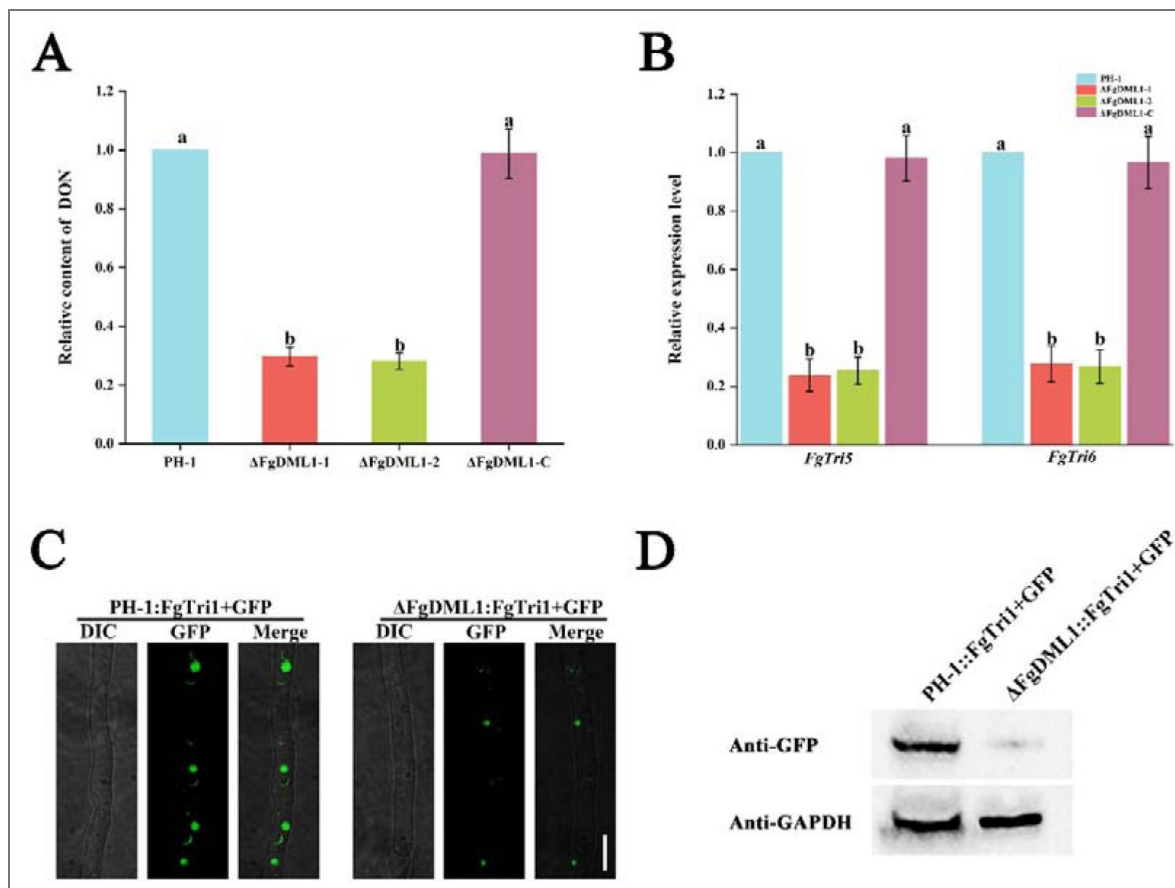


Figure 4. FgDML1 is required for the biosynthesis of DON toxin in *Fusarium graminearum*.

(A) ΔFgDML1 exhibited decreased DON content. All strains were incubated in trichothecene biosynthesis induction (TBI) medium for 7 days. (B) Relative mRNA expression levels of *FgTRI5*, and *FgTRI6* in the tested strains. Mycelia were harvested after 2 days of cultivation in TBI medium, and mRNA were extracted for quantification using the $2^{-\Delta\Delta CT}$ method. *FgGapdh* was used as the reference gene. (C) FgDML1 impaired the formation of DON toxismes, as visualized by Tri1-GFP labeling of toxismes. Strains were cultured in TBI for 36 hours. Bar = 10 μm (D) ΔFgDML1 exhibited reduced expression levels of Tri1-GFP. Strains were cultured in TBI for 36 hours, and mycelia were harvested for western blot analysis. Bars with the same letter indicate no significant difference according to a significant difference (LSD) test at $p = 0.05$.

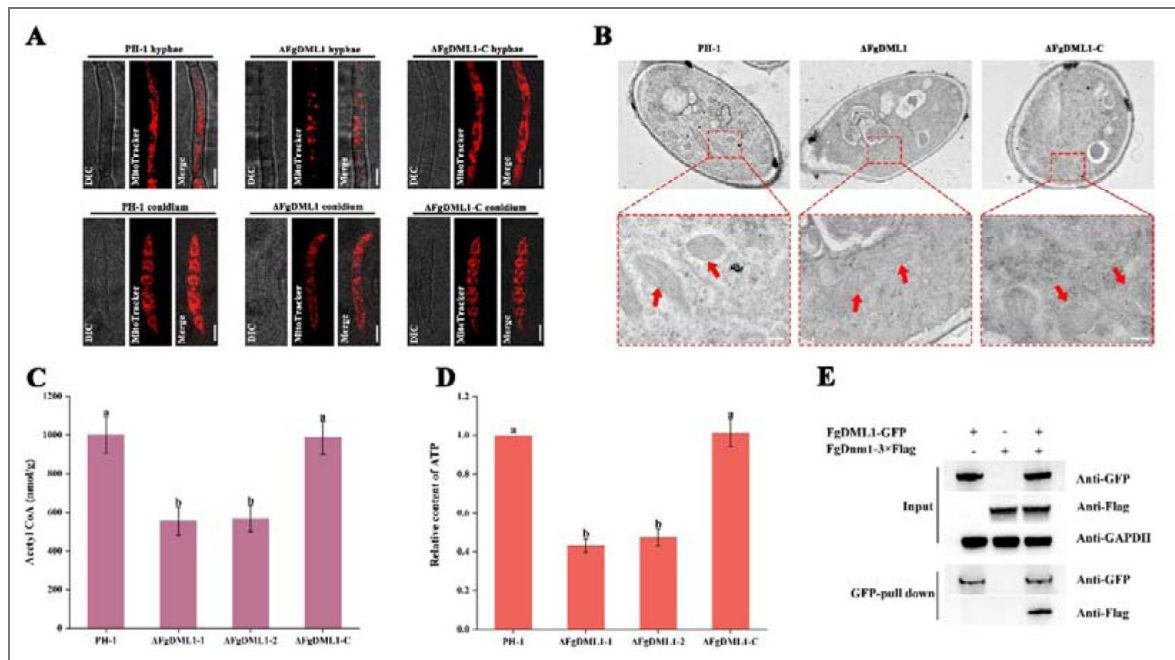


Figure 5. FgDML1 modulates mitochondrial morphology via interaction with FgDnm1 and negatively regulates Acetyl-CoA and ATP synthesis.

(A) Effects of FgDML1 on mitochondrial morphology. Mycelial plugs of the wild-type strain PH-1, the Δ FgDML1 deletion mutant, and the complemented mutant Δ FgDML1-C were cultured in YEPD medium for 36 hours. Mitochondria were stained using a mitochondria-specific fluorescent dye and observed under a confocal microscope. Additionally, conidial germlings were cultured in MBL medium and subjected to the same staining and imaging procedures. Scale bar = 5 μ m. (B) Ultrastructural analysis of mitochondria by transmission electron microscopy (TEM). Mitochondrial ultrastructure was further examined in PH-1, Δ FgDML1, and Δ FgDML1-C using TEM. Mitochondria are indicated by red arrows. Scale bar = 200 nm. (C) FgDML1 Deletion Reduces Acetyl-CoA Levels. Mycelial cultures of PH-1, Δ FgDML1, and Δ FgDML1-C were incubated in YEPD medium for 36 hours. Acetyl-CoA levels were quantified using a commercial assay kit, following the instructions provided by the commercial assay kit. (D) FgDML1 deletion reduces ATP levels. The ATP content of PH-1, Δ FgDML1, and Δ FgDML1-C was measured following the same culture conditions as in (C), using a commercial ATP assay kit. (E) Co-immunoprecipitation (Co-IP) confirms interaction between FgDML1 and FgDnm1. Total protein extracts from strains expressing tagged versions of FgDML1 and FgDnm1 were separated by SDS-PAGE and subjected to immunoblot analysis. Co-IP was performed using ChromoTek GFP-Trap® magnetic agarose to pull down interacting proteins. Immunodetection was carried out using polyclonal anti-Flag and monoclonal anti-GFP antibodies. Monoclonal anti-GAPDH antibody was used as a reference for protein samples. Bars with the same letter indicate no significant difference according to a significant difference (LSD) test at $p = 0.05$.

mitochondrial fission, the interaction between FgDML1 and FgDnm1 was verified through co-immunoprecipitation (CoIP) assays (Fig. 5E [↗](#)). These findings indicate that FgDML1 plays a pivotal role in maintaining mitochondrial morphology by interacting with FgDnm1.

The deletion of FgDML1 reduces Complex III activity, induces compensatory upregulation of FgQCR2, FgQCR8, FgQCR9

Deletion of FgDML1 reduced *F. graminearum* sensitivity to the Qi-site fungicide cyazofamid, while having no effect on sensitivity to the Qo-site inhibitors pyraclostrobin and trifloxystrobin, nor to other respiratory inhibitors such as pydiflumetofen, fluxapyroxad, and fluopyram (Fig. 6A, B [↗](#)) (Fig. S3 [↗](#)). Measurement of Complex III activity in Δ FgDML1 showed a 38.28% reduction in enzyme activity compared to PH-1 (Fig. 6C [↗](#)). To further investigate the reduced sensitivity to cyazofamid, transcription levels of the three subunits (*FgCytb*, *FgCytC1*, *FgISP*) and five assembly factors (*FgQCR2*, *FgQCR6*, *FgQCR7*, *FgQCR8*, *FgQCR9*) of Complex III were analyzed. The results revealed that, upon Cyazofamid treatment, the expression of *FgQCR2*, *FgQCR7*, and *FgQCR8* was significantly upregulated in Δ FgDML1 (Fig. 6D [↗](#)). To further clarify whether the upregulated expression of *FgQCR2*, *FgQCR7*, and *FgQCR8* genes affects their protein expression levels, we measured the protein levels. The results showed that the protein expression levels of FgQCR2, FgQCR7, and FgQCR8 in Δ FgDML1 were higher than those in PH-1 (Fig. 6F [↗](#)). Subsequently, we overexpressed *FgQCR2*, *FgQCR7*, and *FgQCR8* in the wild-type background, and the corresponding overexpression mutants exhibited reduced sensitivity to cyazofamid (Fig. 6E [↗](#)).

Overexpression of FgQCR2, FgQCR8, and FgQCR9 may alters the conformation of the QI site, resulting in reduced sensitivity to cyazofamid

Since Complex III is involved in the action of both cyazofamid (targeting the QI site) and pyraclostrobin (targeting the QO site), the sensitivity of Δ FgDML1 to cyazofamid and pyraclostrobin was investigated. The results showed that the deletion of FgDML1 did not alter sensitivity to a pyraclostrobin (Fig. 6B [↗](#)). Δ FgDML1 was subsequently treated with pyraclostrobin or cyazofamid, and Complex III enzyme activity was measured. After pyraclostrobin treatment, no significant change in enzyme activity was observed in Δ FgDML1 compared to PH-1 and Δ FgDML1-C (Fig. 6C [↗](#)). However, when treated with cyazofamid, a significant increase in Complex III enzyme activity was observed in Δ FgDML1, while the control groups maintained relatively low enzyme activity (Fig. 6C [↗](#)). These findings, combined with the upregulation of assembly factors, suggest that the elevated expression of assembly factors alters the protein conformation at the QI site, modulating sensitivity to cyazofamid.

FgDML1 regulates sensitivity to various stresses

F. graminearum encounters various stresses during infection plant, which significantly influence its ability to infect. To investigate its response to different stress conditions, the sensitivity of Δ FgDML1 to various environmental factors was assessed, including osmotic stress (NaCl, KCl, sorbitol), cell wall stress (Congo red), membrane stress (SDS), as well as high temperature (30°C) and low temperature (15°C) stresses. The results revealed that Δ FgDML1 displayed significantly reduced sensitivity to osmotic stress compared to PH-1 and Δ FgDML1-C (Fig. 7A, B [↗](#)). No significant difference was observed in sensitivity to membrane stress, while a notable change in sensitivity to cell wall stress was observed (Fig. 7C, D [↗](#)). Additionally, differences in response to both high and low temperature stress were noted in Δ FgDML1 when compared to the control strains PH-1 and Δ FgDML1-C (Fig. 7E, F [↗](#)).

Discussion

Misato Protein was originally identified in *Drosophila melanogaster*, where it was reported to play essential roles in mitochondrial DNA inheritance, cell division, and energy metabolism (Miklos et al., 1997 [↗](#)). The Misato/DML1 protein family is evolutionarily conserved from yeast to humans

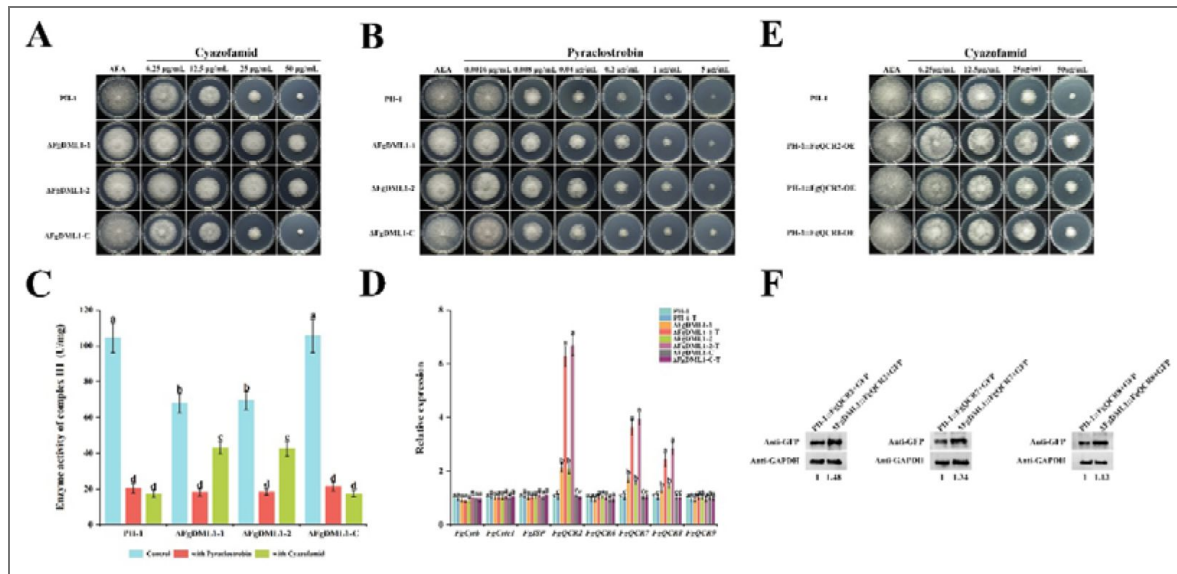


Figure 6. FgDML1 modulates the sensitivity of *Fusarium graminearum* to cyazofamid.

(A) Δ FgDML1 exhibits reduced sensitivity to cyazofamid. PH-1, Δ FgDML1, and Δ FgDML1-C were cultured on AEA medium modified with cyazofamid, with 50 μ g/mL SHAM included to inhibit the alternative oxidase pathway. (B) Deletion of *FgDML1* did not affect *F. graminearum* sensitivity to pyraclostrobin. The tested strains were incubated on AEA medium containing pyraclostrobin and 50 μ g/mL SHAM to assess their response to the fungicide. (C) FgDML1 deletion reduces complex III enzymatic activity. Mycelial cultures of PH-1, Δ FgDML1, and Δ FgDML1-C were grown in YEPD for 36 hours, then treated with 30 μ g/mL cyazofamid, 1.5 μ g/mL pyraclostrobin, and 50 μ g/mL SHAM for an additional 12 hours. The enzymatic activity of complex III was quantified using a commercial assay kit. (D) mRNA expression levels of complex III subunits and assembly factors in PH-1, Δ FgDML1, and Δ FgDML1-C. PH-1, Δ FgDML1, and Δ FgDML1-C were incubated in liquid AEA medium for 36 hours, followed by treatment with 30 μ g/mL cyazofamid and 50 μ g/mL SHAM. The relative mRNA expression levels of complex III subunits and assembly factors were quantified using the $2^{-\Delta\Delta CT}$ method. "T" denotes cyazofamid treatment. Bars with the same letter indicate no significant difference according to a significant difference (LSD) test at $p = 0.05$. (E) Overexpression of *FgQCR2*, *FgQCR7*, and *FgQCR8*, respectively, in the PH-1 background resulted in reduced sensitivity to cyazofamid in the overexpression mutants. (F) Protein expression levels of FgQCR2, FgQCR7, and FgQCR8 in PH-1 and Δ FgDML1. Strains were cultured in AEA Liquid Medium for 36 h, and mycelia were harvested for western blot analysis.

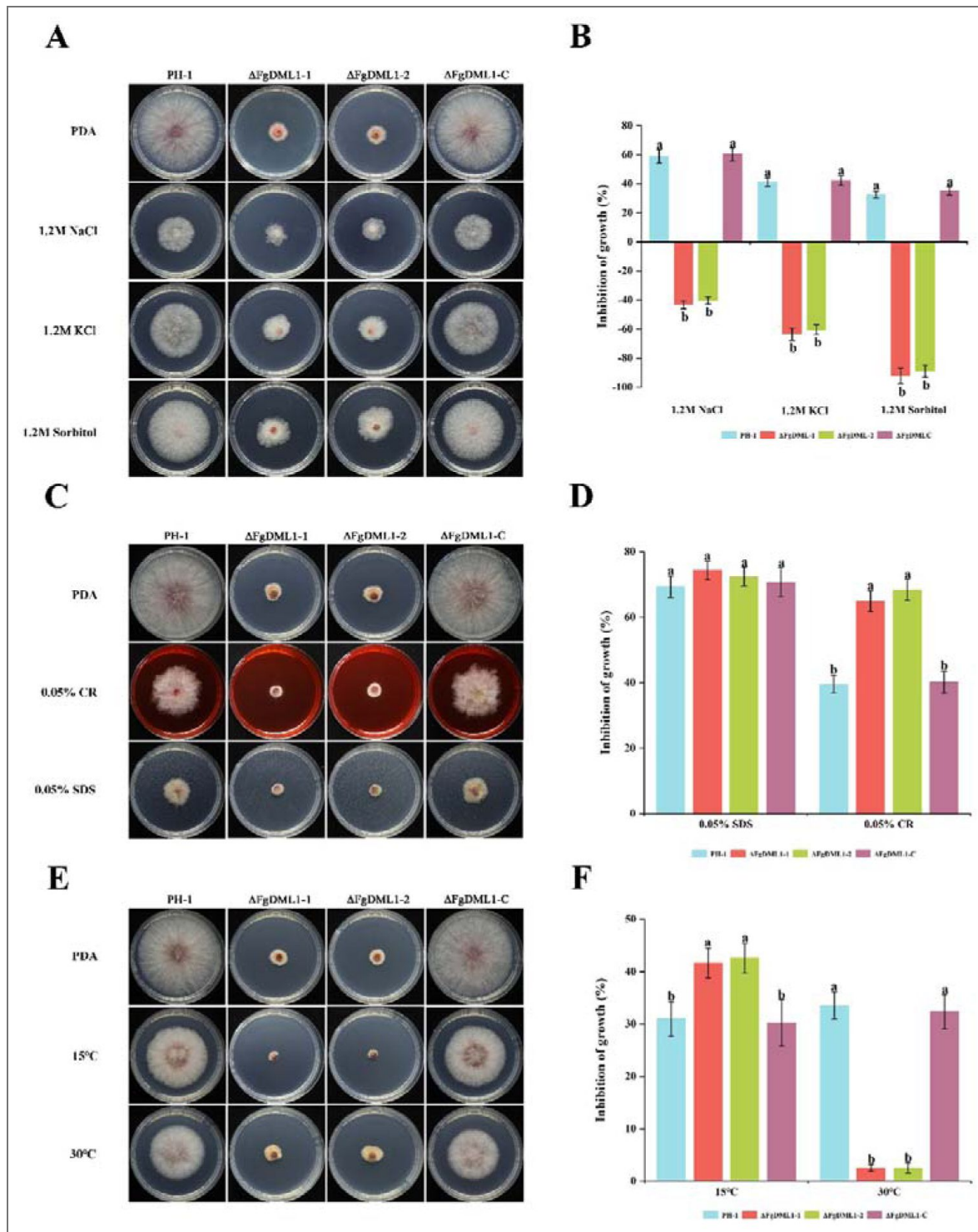


Figure 7. Response of the strains to different stresses.

(A) Reduced sensitivity to osmotic stress in $\Delta FgDML1$. PH-1, $\Delta FgDML1$ and $\Delta FgDML1-C$ were subjected to osmotic stress using NaCl, KCl, and Sorbitol to evaluate their sensitivity. (B) Statistical analysis of growth inhibition under osmotic stress. Growth inhibition rates of all strains under NaCl, KCl, and Sorbitol stress were statistically analyzed. (C) Sensitivity to cell wall and membrane damaging agents. The tested strains were exposed to cell wall-damaging agent Congo Red (CR) and cell membrane-damaging agent Sodium dodecyl sulfate (SDS) to assess their sensitivity. (D) Statistical analysis of growth inhibition under CR and SDS stress. Growth inhibition rates of all strains under CR and SDS stress were statistically analyzed. (E) Sensitivity to temperature stresses. The sensitivity of the tested strains to different temperature stresses was evaluated. (F) Statistical analysis of growth inhibition under temperature stress. Growth inhibition rates of all strains under various temperature stresses were statistically analyzed. Significant differences were determined using the LSD test at $p = 0.05$. Bars with the same letter indicate no significant difference.

and plays a critical role in mitochondrial regulation. In *S. cerevisiae*, *DML1* is an essential gene; its deletion is lethal, while its overexpression results in fragmented mitochondrial networks and aberrant cellular morphology, underscoring its necessity for normal mitochondrial function (Gurvitz et al., 2002 [\[4\]](#)). Similarly, in *Homo sapiens*, the homolog Misato localizes to the mitochondrial outer membrane, and both its depletion and overexpression are sufficient to disrupt mitochondrial morphology and distribution (Kimura and Okano, 2007 [\[5\]](#)). However, its functional characterization in filamentous fungi remains largely unexplored. Our study defines a novel mechanism through which FgDML1 governs mitochondrial homeostasis. We demonstrate that FgDML1 directly interacts with the key mitochondrial fission regulator FgDnm1 and positively modulates cellular bioenergetic metabolism, as evidenced by elevated ATP and acetyl-CoA levels (Fig. 8 [\[4\]](#)). In this study, we identified and analyzed the homolog FgDML1 in the phytopathogenic fungus *F. graminearum*. The FgDML1 protein contains conserved Misato_Tub_SegII and Tubulin_3 domains, with high homology observed among species within the *Fusarium* genus but significant genetic divergence from other taxa (Fig. 1A [\[4\]](#)). In *S. cerevisiae*, DML1 is a critical regulator of mitochondrial fission and fusion dynamics, maintaining intracellular energy homeostasis by modulating mitochondrial morphology and distribution (Gurvitz et al., 2002 [\[4\]](#)). Our study demonstrated that FgDML1 localized to mitochondria in *F. graminearum* (Fig. 1B [\[4\]](#)). Deletion mutants of *FgDML1* exhibited phenotypes indicative of mitochondrial dysfunction, including fragmented mitochondrial morphology (Fig. 5A [\[4\]](#)). TEM further confirmed structural and dynamic damage to mitochondria in the Δ FgDML1 (Fig. 5B [\[4\]](#)). Structural analysis suggests that DML1 shares similarities with GTPase family proteins, indicating potential GTPase activity (Gurvitz et al., 2002 [\[4\]](#)). GTPases primarily hydrolyze GTP, but their activity is intricately linked to ATP production and utilization. These enzymes regulate critical cellular processes such as energy metabolism, signal transduction, protein synthesis, and cytoskeletal remodeling, all of which influence ATP levels (Mrnjavac and Martin, 2025 [\[4\]](#); Pb et al., 2023 [\[4\]](#); Rogne et al., 2018 [\[4\]](#)). In agreement with this, we observed a significant reduction in ATP content in the FgDML1 deletion mutant, underscoring its role in mitochondrial energy metabolism (Fig. 5D [\[4\]](#)). In *H. sapiens*, DML1 also localizes to mitochondria, where its silencing results in fragmented and aggregated mitochondria with reduced network continuity and fusion activity (Kimura and Okano, 2007 [\[5\]](#)). Our findings indicate that FgDML1 plays a conserved role in maintaining mitochondrial integrity and energy homeostasis. Notably, FgDML1 was found to interact with FgDnm1 (Fig. 5E [\[4\]](#)). FgDnm1 is a key dynamin-related protein mediating mitochondrial fission (Griffin et al., 2005 [\[4\]](#); Kang et al., 2023 [\[4\]](#)), suggesting that FgDML1 may form a complex with FgDnm1 to regulate mitochondrial fission and fusion processes. To our knowledge, this is the first report documenting an interaction between DML1 and Dnm in any fungal species, including model organisms such as *S. cerevisiae*. This novel finding provides new insights into the molecular mechanisms underlying mitochondrial dynamics in filamentous fungi.

The silencing of ATP synthase subunit α (AtpA) has been shown to reduce ATP synthesis and inhibit the biosynthesis of TcdA and TcdB toxins (Marreddy et al., 2024 [\[4\]](#)). In plants, the production of secondary metabolites often demands substantial energy, with ATP serving as the primary energy carrier within cells. When ATP levels are depleted, plants may prioritize growth and development by reducing the synthesis of certain secondary metabolites, including toxins (Xiao et al., 2024 [\[4\]](#)). Similarly, in *F. graminearum*, deoxynivalenol (DON) toxin, a highly harmful mycotoxin and secondary metabolite, requires sufficient energy for its biosynthesis. In our study, the ATP content in the FgDML1 deletion mutant was significantly lower, and DON production was markedly reduced, compared to the wild-type PH-1 and the complementary mutant Δ FgDML1-C (Fig. 4A [\[4\]](#)). Reduced ATP levels disrupt the formation of toxisomes, which are critical for DON toxin biosynthesis (Tang et al., 2018 [\[4\]](#)), and the Δ FgDML1 strain is unable to form functional toxisomes (Fig. 4C [\[4\]](#)). Western blot analysis also confirmed a significant reduction in the protein expression of FgTri1 (Fig. 4D [\[4\]](#)). DON production is predominantly regulated by *Tri* genes, with *Tri5* and *Tri6* serving as critical components in the biosynthetic pathway (Kimura et al., 2001 [\[4\]](#)). *Tri5* encodes trichodiene synthase, the key enzyme catalyzing the first step of DON biosynthesis (Huang et al., 2024 [\[4\]](#); Jiang et al., 2016 [\[4\]](#)). In our study, the expression levels of *FgTri5* and *FgTri6* were significantly downregulated in the FgDML1 deletion mutant (Fig. 4B [\[4\]](#)). To our

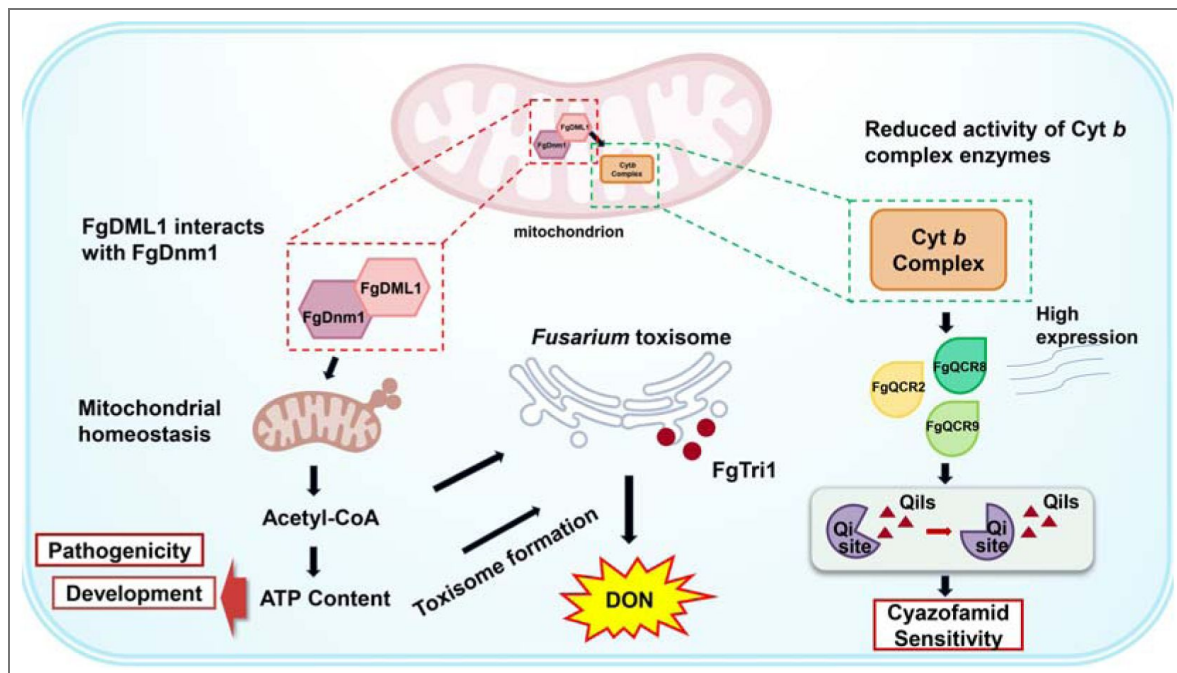


Figure 8. Model of FgDML1 regulation of *Fusarium graminearum* sensitivity to DON toxin synthesis and cyazofamid fungicide.

FgDML1 interacts with FgDnm1 to modulate mitochondrial morphology, thereby positively regulating acetyl-CoA and ATP levels, which influence toxosome formation and ultimately affect DON production. Loss of FgDML1 reduces complex III enzymatic activity, leading to the upregulation of the assembly factors *FgQCR2*, *FgQCR8*, and *FgQCR9*. This alteration in assembly factor expression changes the conformation of the Qi site, leading to reduced sensitivity to cyazofamid.

knowledge, no research has yet elucidated a regulatory role for DML in gene expression in either *S. cerevisiae* or *D. melanogaster*. However, given its established functions in cell division and microtubule organization, FgDML1 might indirectly influence gene expression by modulating chromatin structure or cell cycle progression (Schulze and Wallrath, 2007). Investigating this hypothesis will be a key focus of our future research. Given this substantial reduction in DON production, we investigated whether the precursor for DON biosynthesis, acetyl-CoA, was also affected. Results revealed a significant decrease in acetyl-CoA levels in the Δ FgDML1 mutant (Fig. 5C). In conclusion, reduced ATP levels impair toxin body formation, leading to lower toxin production. Additionally, decreased acetyl-CoA disrupts the availability of essential precursors for DON toxin synthesis, further decreasing its production. The combined effects of these factors result in a significant reduction in DON toxin biosynthesis.

F. graminearum is a highly destructive pathogen that poses a significant threat to wheat production worldwide. Cyazofamid is an effective fungicide commonly used to control plant pathogenic fungi. Cyazofamid specifically targets the Qi site of complex III in the mitochondrial respiratory chain (Bolgunas et al., 2006; Mitani et al., 2001). Complex III, also known as the cytochrome bc1 complex, plays a critical role in the electron transport chain by transferring electrons from ubiquinone to cytochrome c and pumping protons from the mitochondrial matrix into the intermembrane space (Fernández-Ortuño et al., 2008; Sarewicz et al., 2013). This complex consists of several core subunits, including cytochrome b, cytochrome c1, and iron-sulfur proteins, as well as multiple assembly factors. It contains two quinone binding sites: the Qo site (quinone oxidation site) and the Qi site (quinone reduction site) (Vercellino and Sazanov, 2022; Zara et al., 2009). Numerous fungicides have been developed to inhibit the Qo site (e.g., pyraclostrobin, azoxystrobin) (Nuwamanya et al., 2022; Peng et al., 2022) and the Qi site (e.g., cyazofamid) (Mitani et al., 2001) of the cytochrome bc1 complex. In our study, we observed that the deletion of *FgDML1* significantly reduced the sensitivity of *F. graminearum* to cyazofamid (Fig. 6A), while its sensitivity to pyraclostrobin remained unaffected (Fig. 6B). In the Δ FgDML1 mutant, we detected a decrease in complex III enzyme activity (Fig. 6C) and noted the upregulation of assembly factors *FgQCR2*, *FgQCR7*, and *FgQCR8* following cyazofamid treatment (Fig. 6D). Further enzyme activity assays revealed that after pyraclostrobin treatment, Δ FgDML1 exhibited a 73.06% reduction in complex III activity, compared to an 80.38% decrease in the wild-type strain PH-1 (Fig. 6C). Interestingly, after cyazofamid treatment, the decrease in enzyme activity in Δ FgDML1 was only 36.31%, while PH-1 showed a dramatic 83.46% reduction in complex III activity. *FgQCR2*, *FgQCR7*, and *FgQCR8* are critical assembly factors that play essential roles in the proper assembly of complex III (Zhan et al., 2024). These factors stabilize the core subunit structure, promoting correct folding and functional operation of complex III, particularly during the assembly of the Qi site. This ensures the proper formation and function of the Qi site, thus supporting the normal operation of the mitochondrial respiratory chain (Wang et al., 2024; Zara et al., 2007). Previous studies have reported that mutations in the Complex III assembly factors TTC19, UQCC2, and UQCC3 impair the assembly and activity of Complex III (Feichtinger et al., 2017; Wanschers et al., 2014). Given the role of FgDML1 in mitochondrial homeostasis, we further evaluated the sensitivity of the Δ FgDML1 mutant to other commonly used respiratory chain inhibitors, including pydiflumetofen, fluxapyroxad, fluopyram, and trifloxystrobin. The results indicated that FgDML1 deletion had no impact on the sensitivity of *F. graminearum* to these fungicides (Fig. S3). In conclusion, our findings suggest that the overexpression of assembly factors *FgQCR2*, *FgQCR7*, and *FgQCR8* in Δ FgDML1 potentially modifies the conformation of the Qi site, which specifically modulates the sensitivity of *F. graminearum* to cyazofamid.

In *H. sapiens*, mutations or deletions in DML1 are associated with mitochondrial diseases characterized by early-onset myopathy and cerebellar ataxia (Chen et al., 2022; Nasca et al., 2017). In our study, the deletion of *FgDML1* in *F. graminearum* resulted in significant defects in vegetative growth, including a markedly reduced growth rate (Fig. 2A) and an aberrant, excessively curved hyphal morphology (Fig. 2B). Notably, the loss of FgDML1 rendered *F. graminearum* incapable of completing sexual reproduction (Fig. 2D), thereby potentially reducing its primary inoculum sources. Additionally, the deletion of FgDML1 impaired asexual

reproduction and pathogenicity (Fig. 2 [↗](#)) (Fig. 3 [↗](#)), critical factors for fungal survival and host infection. Furthermore, the Δ FgDML1 mutant exhibited altered responses to various stress conditions (Fig. 7 [↗](#)), suggesting a broader role for FgDML1 in regulating fungal stress tolerance. In the present study, FgDML1 is localized to the mitochondria and interacted with FgDnm1, a key regulator of mitochondrial morphology, positively regulates ATP and acetyl-CoA levels, influences toxin body formation, and ultimately leads to a significant reduction in DON toxin biosynthesis. Furthermore, deletion of FgDML1 impairs the enzymatic activity of Complex III, which subsequently induces the upregulation of the assembly factors FgQCR2, FgQCR8, and FgQCR9, resulting in conformational changes at the Qi site of Complex III and decreased sensitivity to cyazofamid. In addition, the absence of FgDML1 has profound effects on pathogenicity, fungal development, and mitochondrial morphology. Based on these findings, we propose a working model delineating the role of FgDML1 in regulating DON toxin biosynthesis and cyazofamid sensitivity in *F. graminearum* (Fig. 8 [↗](#)). This study not only provides a theoretical basis for understanding the regulatory mechanisms underlying toxin biosynthesis in *F. graminearum* but also identifies potential molecular targets for the development of novel agents to control FHB.

Material and methods

Strains, culture conditions, and fungicides

The wild-type *F. graminearum* strain PH-1 was used as the parental strain for protoplast transformation experiments. All strains were preserved at -80°C. The tested strains were cultured in the dark on potato dextrose agar (PDA) at 25°C. To evaluate vegetative growth, complete medium (CM), minimal medium (MM), and V8 Juice Agar (V8) media were prepared as described previously (Tang et al., 2020 [↗](#)). Mung bean liquid (MBL) medium was used for conidial production, while carrot agar (CA) medium was utilized to assess sexual reproduction (Wang et al., 2011 [↗](#)). DON toxin production was measured using trichothecene biosynthesis-inducing (TBI) medium (Wang et al., 2024 [↗](#)).

Cyazofamid (98%, BASF China Ltd.), pyraclostrobin (98%, BASF China Ltd.) pydiflumetofen (98%, China Syngenta crop Protection Co., Ltd.), trifloxystrobin (96%, Bayer Crop Science China Co., Ltd.), fluxapyroxad (98%, BASF China Ltd.) and fluopyram (96%, Bayer Crop Science China Co., Ltd.) were dissolved in dimethyl sulfoxide (DMSO) at stock concentrations of 50,000 µg/mL and 10,000 µg/mL, respectively.

Generation of deletion mutants and complemented mutants

The split-marker method was employed to generate the Δ FgDML1 mutant (Yu et al., 2004 [↗](#)). Specifically, approximately 1.2 kb of the upstream and downstream flanking regions of FgDML1 were amplified and fused with a dual-selection marker containing the hygromycin phosphotransferase (*HPH*) gene and the herpes simplex virus thymidine kinase (*HSV-tk*) gene fragments. The resulting fusion fragment was transformed into the wild-type *F. graminearum* PH-1 strain via polyethylene glycol (PEG)-mediated protoplast transformation. Transformants were selected on PDA plates containing either 100 µg/mL Hygromycin B (Yeasen, Shanghai, China) or 0.2 µmol/mL 5-Fluorouracil 2'-deoxyriboside (F2du) (Solarbio, Beijing, China) (Zhao et al., 2022 [↗](#)). Positive transformants were initially screened by PCR and subsequently verified through Southern blot analysis. To generate the complemented mutant Δ FgDML1-C, the full-length coding sequence (CDS) of FgDML1 was introduced into the Δ FgDML1 mutant, replacing the dual-selection cassette. Furthermore, FgDML1-GFP and FgDnm1-3×Flag mutants were constructed using previously established methods for further functional studies (Wu et al., 2022 [↗](#)). Specifically, FgDML1, including its native promoter region and open reading frame (ORF) (excluding the stop codon), was amplified. The PCR product was then fused with the *Xho*I-digested pYF11 vector. After transformation into *E. coli* and sequence verification, the plasmid was extracted and subsequently introduced into PH-1 protoplasts. For FgDnm1-3×Flag, the 3×Flag tag was added to the C-terminus of FgDnm1 by PCR, fused with the hygromycin resistance gene and the FgDnm1 downstream arm, and then introduced into PH-1 protoplasts. The overexpression mutant was constructed according

to a previously described method. Specifically, the ORF of FgDML1 was amplified and the PCR product was ligated into the *Sac*II-digested pSXS overexpression vector. The resulting plasmid was then transformed into PH-1 protoplasts (Shi et al., 2023 [DOI](#)). For the construction of PH-1::FgTri1+GFP and Δ FgDML1::FgTri1+GFP, the ORF of FgTri1 was amplified and ligated into the *Xho*I-digested pYF11 vector as described above. The resulting vectors were then transformed into protoplasts of PH-1 or Δ FgDML1, respectively.

Vegetative growth and conidiation assays

Fresh mycelial plugs of the tested strains were transferred to PDA, CM, MM, or V8 plates to evaluate mycelial growth rates. Conidial production was induced by incubating the strains in MBL medium, and the number of conidia was determined using a hemocytometer. The septa of conidia were visualized through calcofluor white (CFW) staining. For germination rate assessment, conidia were collected and inoculated onto water agar (WA) plates, followed by incubation at 25°C for 2 and 8 hours. Germination rates were then recorded at the respective time points. For each sample, a total of 200 conidia were counted. The experiment included three biological replicates with three technical replicates each.

Sexual reproduction assay

Sexual reproduction experiments were performed on CA medium. Test strains were incubated on CA medium in complete darkness for 6 days. After incubation, the surface mycelia were removed using a mycelial scraper and 2.5% Tween 60 solution. The plates were then further incubated at 25°C under a 12-hour light/12-hour dark cycle to induce sexual development. Perithecia and ascospores were observed and recorded on days 7 and 14 using a Nikon SMZ25 fluorescent stereomicroscope and an Olympus IX-71 inverted fluorescent microscope. Each treatment group contains three biological replicates.

Microscopic observation

To determine the subcellular localization of FgDML1, the FgDML1-GFP mutant was incubated in YEPD medium at 25°C with shaking at 145 × g for 36 hours. Subcellular localization was analyzed using a Leica TCS SP8 confocal microscope, with MitoTracker Red CMXRos (Yeasen, Shanghai, China) applied to confirm mitochondrial targeting through colocalization of fluorescence signals. To investigate toxisome morphology, the Δ FgDML1::Tri1+GFP mutant was cultured in TBI medium at 28°C with shaking for 36 hours, and the structural dynamics of toxisomes within the mycelium were visualized using Tri1-GFP fluorescence under confocal microscopy. Additionally, mitochondrial morphology was evaluated by culturing PH-1, Δ FgDML1, and Δ FgDML1-C in YEPD medium under identical conditions, followed by staining with MitoTracker Red CMXRos and confocal microscopy analysis to assess mitochondrial structural alterations, distribution, and dynamics. For ultrastructural observations, mycelial samples were fixed and dehydrated according to standard protocols, embedded, sectioned, and stained with uranyl acetate and lead citrate. Mitochondrial ultrastructure was examined using a Tecnai G2 Spirit Bio electron microscope (Thermo, MA, USA) at an accelerating voltage of 80 kV to compare the structural integrity and morphological differences among the strains. Each treatment group contains three biological replicates and three technical replicates.

Determination of DON production and pathogenicity

As previously described (Tang et al., 2020 [DOI](#); Wang et al., 2025 [DOI](#)), Specifically, coleoptiles were inoculated with conidial suspensions and incubated for 14 days, while leaves were inoculated with fresh mycelial plugs and incubated for 5 days, followed by observation and quantification of disease symptoms. DON toxin was measured using a Wise Science ELISA-based kit (Wise Science, Jiangsu, China) (Li et al., 2019 [DOI](#); Zheng et al., 2018 [DOI](#)). Fresh mycelial plugs of the test strains were placed in TBI medium under toxin-producing conditions (28°C, 145 × g, 7 days incubation), and

DON toxin was quantified according to the commercial kit instructions after harvesting mycelia and toxin-containing filtrates. Each treatment group contains three biological replicates and three technical replicates.

Determination of acetyl-CoA content and ATP concentration

ATP content and acetyl-CoA levels were determined using the methods described previously (Hu et al., 2023 [\[1\]](#); Song et al., 2024 [\[2\]](#)). The test strains were cultured in yeast extract peptone dextrose (YEPD) medium at 25°C and 145 × g for 36 hours. Mycelia were collected by filtration through three layers of lens-cleaning paper, and excess moisture was removed with filter paper. Subsequently, ATP and acetyl-CoA levels were quantified using ATP and acetyl-CoA assay kits (Solarbio, Beijing, China), respectively. Each treatment group contains three biological replicates and three technical replicates.

Sensitivity determination

To investigate the sensitivity of ΔFgDML1 to QoI and QiI fungicides, the wild-type strain PH-1, ΔFgDML1 mutant, and the complemented mutant ΔFgDML1-C were inoculated on AEA plates supplemented with gradient concentrations of cyazofamid and pyraclostrobin. The wild-type PH-1 and its overexpression mutants (PH-1::FgQCR2-OE, PH-1::FgQCR7-OE, and PH-1::FgQCR8-OE) were assayed on AEA plates amended with cyazofamid and supplemented with SHAM. All strains were incubated at 25°C in darkness; however, due to ΔFgDML1 slower growth, the ΔFgDML1 mutant required a 5-day incubation period compared to the 3 days used for PH-1 and ΔFgDML1-C. The concentration gradients for each fungicide in the sensitivity assays were set up according to Supplementary Table S2 [\[3\]](#). To inhibit the alternative respiratory pathway, 50 μg/mL SHAM was applied. The percentage inhibition (%) of mycelial growth was calculated using the formula: (mean control diameter - mean diameter at fungicide concentration) / (mean control diameter - 5 mm mycelial plug) × 100. To evaluate the sensitivity of ΔFgDML1 to various stress conditions, the test strains were cultured on PDA plates supplemented with stress-inducing agents, including NaCl, KCl, sorbitol, Congo red, and SDS. Plates were incubated at 25°C for 3 days, and colony growth was assessed. Each treatment group contains three biological replicates.

Measurement of complex III enzyme activity

The enzyme activity of Complex III was determined in the PH-1, ΔFgDML1, and ΔFgDML1-C. Fresh mycelial plugs from each strain were inoculated into liquid AEA medium and incubated at 25°C with shaking at 145 × g for 36 hours. Subsequently, cyazofamid and pyraclostrobin were added to final concentrations of 30 μg/mL and 1.5 μg/mL, respectively, along with 50 μg/mL SHAM. After an additional 12-hour incubation, the mycelia were harvested, and Complex III activity was measured according to the complex III enzyme activity kit (Solarbio, Beijing, China) instructions. Briefly, 0.1 g of mycelia was homogenized with 1 mL of extraction buffer in an ice bath. The homogenate was centrifuged at 600 × g for 10 min at 4°C. The resulting supernatant was then subjected to a second centrifugation at 11,100 × g for 10 min at 4°C. The pellet was resuspended in 200 μL of extraction buffer and disrupted by ultrasonication (200 W, 5 s pulses with 10 s intervals, 15 cycles). Complex III enzyme activity was finally measured by adding the working solution as per the manufacturer's protocol. Each treatment group contains three biological replicates and three technical replicates.

RNA extraction and RT-qPCR assays

The test strains were transferred to liquid AEA medium and incubated at 25°C for 48 hours, with cyazofamid added to a final concentration of 20 μg/mL. The control group was not treated with any fungicide. To measure the transcriptional levels of genes involved in DON biosynthesis, the test strains were transferred to TBI medium and incubated for 3 days. Mycelium was collected and total RNA was extracted following the instructions provided by the Total RNA Extraction Kit (Tiangen, Beijing, China). cDNA synthesis was performed using Evo M-MLV Mix Kit (Accurate, Hunan, China). reverse transcription. Reverse Transcription Quantitative Polymerase Chain

Reaction (RT-qPCR) was carried out using the QuantStudio 6 Flex real-time PCR system (Thermo, Fisher Scientific, USA) to assess the relative expression of three subunits of Complex III (*FgCytb*, *FgCyt1*, *FgISP*), five assembly factors (*FgQCR2*, *FgQCR6*, *FgQCR7*, *FgQCR8*, *FgQCR9*), and DON biosynthesis-related genes (*FgTri5* and *FgTri6*). The *FgCox1* gene was used as the reference gene for *FgCytb*, while *FgGapdh* was used as the reference for the other genes. Relative transcript levels were calculated using the $2^{-\Delta\Delta CT}$ method. Each treatment group contains three biological replicates and three technical replicates.

Western Blot assay and Co-IP assay

Total protein extraction was performed according to previously established protocols, and the extracted proteins were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). Subsequently, proteins were transferred onto an Immobilon-P transfer membrane (Millipore, Billerica, USA). The membrane was probed with GFP antibody D191040 (BBI, Shanghai, China) and Flag antibody 30501ES60 (Yeasen, Shanghai, China). The FgDnm1-3×Flag fragment was introduced into PH-1 and FgDML1+GFP protoplasts, respectively, to obtain single-tagged and double-tagged strains. All tagged mutant strains were subjected to triple validation, including PCR, sequencing, and Western blot analysis. Total protein from tagged strains was incubated with ChromoTek GFP-Trap® Magnetic Agarose (ChromoTek, Planegg-Martinsried, Germany) beads. Proteins eluted from the beads were further co-incubated with GFP or Flag antibodies for detection. Gapdh antibody 60004-1-Ig (Proteintech, Rosemont, USA) was used as an internal control for protein normalization. Chemiluminescent detection was performed following incubation with the corresponding secondary antibody. All antibodies were diluted as follows: primary antibodies at 1:1000 and secondary antibodies at 1:10000. The experiment was independently repeated three times.

Statistical analysis

All data were analyzed using one-way analysis of variance (ANOVA), followed by the least significant difference (LSD) test with a significance threshold of $p < 0.05$.

The EC₅₀ values were calculated using DPS statistical analysis software. Each experiment was independently repeated three times.

Supporting Information

Acknowledgements

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Additional information

Data availability

Data will be made available on request.

CRedit authorship contribution statement

Chenguang Wang: methodology, investigation, editing, and writing original draft. Xuwei Mao: formal analysis, data curation. Weiwei Cong, Lin Yang.: performing the partial work of the experiments and analyzing the data. Yiping Hou: conceptualization, resources, writing-reviewing, supervision, funding acquisition.

Fig. S1. Schematic drawing of the generation of Δ FgDML1 and Fg DML1-C.

(A) The deletion of FgDML1. The Δ FgOs1 were constructed via a homologous replacement strategy. Primer pairs P1/P2 and P3/P4 were used to amplify the flanking fragments about 1.2kb upstream and downstream of FgDML1. Primer pairs P5/P6 used to amplify resistance gene hygromycin phosphotransferase (*hph*) and herpes simplex virus-thymidine kinase (*hsv-tk*) fragments. Primer pairs P11/P12 was used to specifically amplify the partial fragment of FgDML1. Primer pairs P7/P8 and P9/P10 were used to verified the upstream and downstream binding fragments, respectively. (B) Generation of the complement strain Δ FgDML1-C. Primer pairs P1/P4 was used to amplify a fragment including the FgDML1 gene with the flanking fragments of FgDML1.

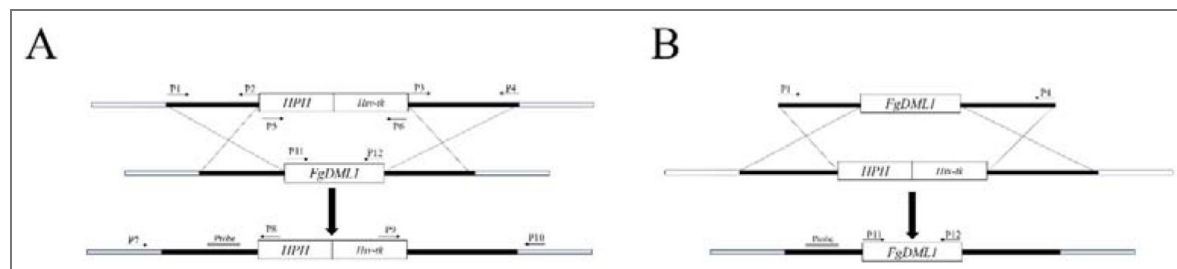
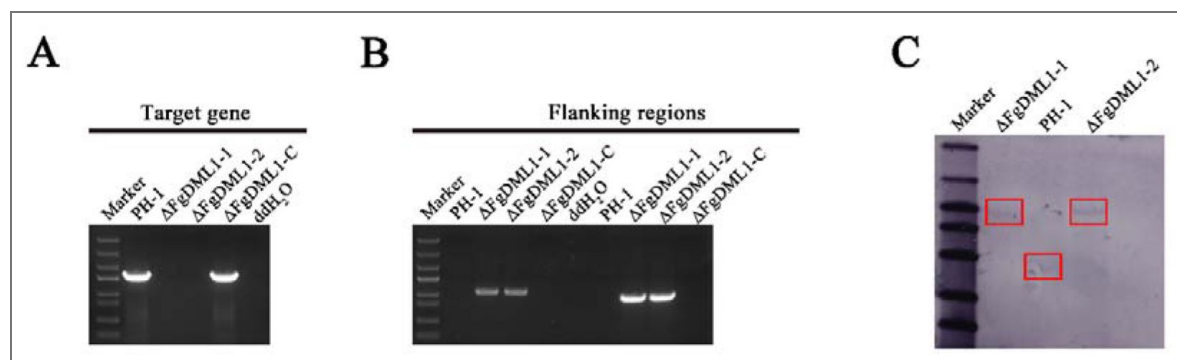


Fig. S2. Identification of Δ FgDML1 and Δ FgDML1-C mutants.

(A) PCR analysis of the target gene for Δ FgDML1 and Δ FgDML1-C. (B) PCR analysis of the flanking regions for Δ FgDML1 and Δ FgDML1-C. (C) Southern blot analysis of PH-1, Δ FgDML1, DNA of the strains was digested by *Cla* I, Probe about 575 bp was located in upstream of Δ FgDML1.



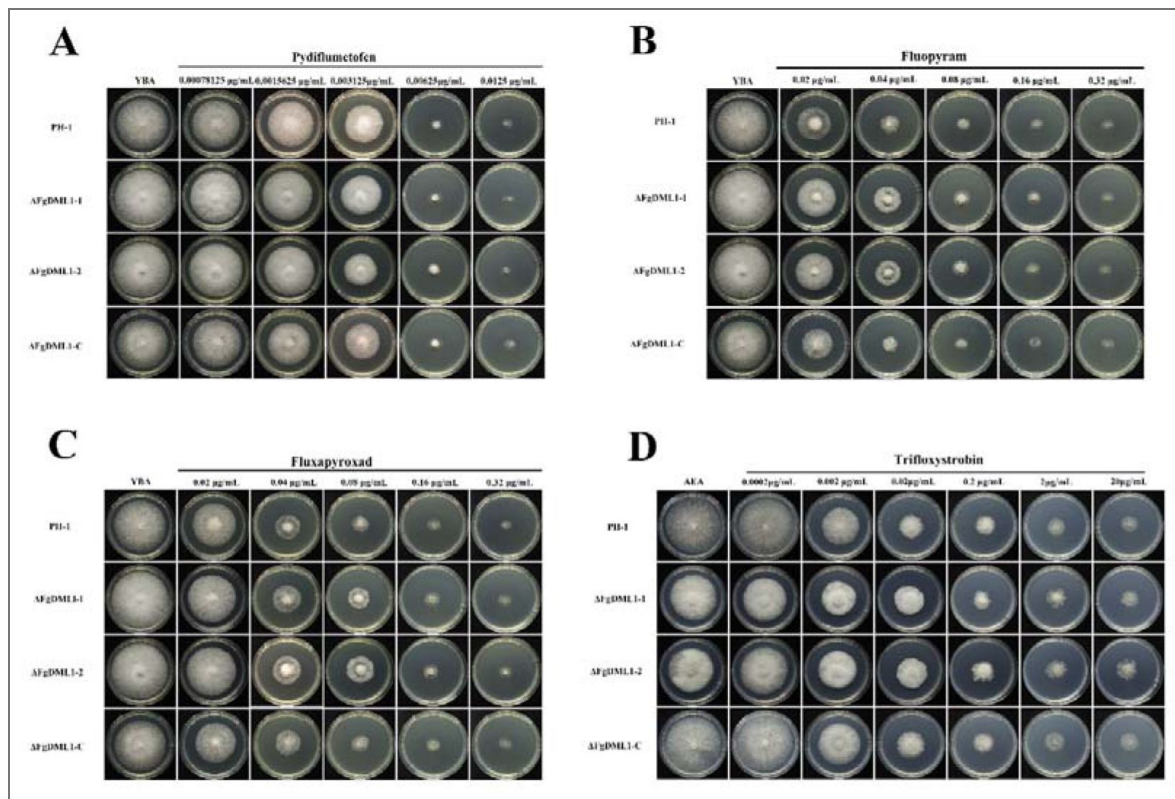


Fig. S3. Sensitivity analysis of Δ FgDML1 to commonly used respiratory chain inhibitors.

(A) Δ FgDML1 shows no change in sensitivity to pydiflumetofen. The tested strains were incubated for 3 days at 25°C on YBA plates supplemented with pydiflumetofen. (B) Δ FgDML1 shows no change in sensitivity to fluxapyroxad. The tested strains were incubated for 3 days at 25°C on YBA plates supplemented with fluxapyroxad. (C) Δ FgDML1 shows no change in sensitivity to fluopyram. The tested strains were incubated for 3 days at 25°C on YBA plates supplemented with fluopyram. (D) Δ FgDML1 shows no change in sensitivity to trifloxystrobin. The tested strains were incubated on AEA plates supplemented with trifloxystrobin and 50 μ g/mL SHAM at 25°C for 3 days.

Table. S1. Primers used in this study

Primers	Sequence (5'→3')	Description
P1	AAACGTGTCTGGCTATAATCATATCG	Upstream fragment of <i>FgDML1</i>
P2	CAAAAAATGCTCTTCAATATCATCTTCTGTAGTCCTTTTGATG GTCAAAGTTCTC	
P3	AACACGGAAGGAGACAATACCGGAAGGAACGAGTGGAATAGTTA ATCGAGAACCACAC	
P4	GGAACTACCACTATCAAGATCGAG	Downstream fragment of <i>FgDML1</i>
P5	ACAGAAGATGATTTGAAGGAGC	
P6	GTTCCTCCGGTATTGTCTC	
P7	ACACTACTAGTGTAGGCCTTCC	Amplify <i>HPH+Hsv-tk</i> sequence
P8	TCATTGGATGCTTGGGTAGAA	
P9	GGCTGCTTGCCAATACGGT	
P10	CATTACCTGGATGTGCAATTTC	Amplify upstream fragment of transformant
P11	ATGCGAGAAATCGTCACCT	
P12	CTATAATCATCGTCGCTCCTCGTCAC	
P13	GGCGGAGGAACACTACTAGTG	Amplify downstream fragment of transformant
P14	CTTAAATGACAACCAGAATGTCTCG	
FgTri5-qPCR-F	GAGTGTTCATGCATGGCTACGTC	
FgTri5-qPCR-R	CTGAGCCTCCTTCACATCGTCC	qPCR for <i>FgTri5</i>
FgTri6-qPCR-F	TATCGAAAATTATATAACCATC	qPCR for <i>FgTri6</i>
FgTri6-qPCR-R	CTGAGGGCATTCTGAGTAGCGACA	
FgGapdh-qPCR-F	CTTACTGCCTCCACCAACTG	
FgGapdh-qPCR-R	TGACGTTGGAAGGAGCGAAG	qPCR for <i>Fggapdh</i>
ISP-qPCR-F	CGAGGGCAAGAACGTTATCA	qPCR for <i>FgISP</i>
ISP-qPCR-R	GGGATCTCGGAGAGAAGAGAT	
Cytb-qPCR-F	GTATCGCAATGTTAGCTGCAA	
Cytb-qPCR-R	CCGCCTTGCTTAAAGGTCTA	qPCR for <i>FgCytb</i>
Cyt1-qPCR-F	CTCTCAGATGGCCAAGGATG	
Cyt1-qPCR-R	GGAGGCAAGACAACAAGAA	
COX1-qPCR-F	AGGGTTGAGTGGTGTGTG	qPCR for <i>FgCox1</i>
COX1-qPCR-R	GCGAATACAGCTCCCATCTT	
QCR2-qPCR-F	CAGCCAGGAGGATATCAAGAAG	
QCR2-qPCR-R	GCAAGAGCAGCAGATCATA	qPCR for <i>FgQCR2</i>
QCR6-qPCR-F	ATGAGGATGACGAGGAGGAA	
QCR6-qPCR-R	TGAACACGCTCAACACACT	
QCR7-qPCR-F	CCTCAAGCGCTCTCTAAC	qPCR for <i>Fg QCR7</i>
QCR7-qPCR-R	GATGTAGTCCACTGGTCTTG	
QCR8-qPCR-F	ACCAAGGGCCAAATCTCTT	
QCR8-qPCR-R	CAGCCTTGGAGTTAAGGTACTC	qPCR for <i>Fg QCR8</i>
QCR9-qPCR-F	GATTGGCTTCAACAACGTCAT	
QCR9-qPCR-R	TTATTCTCTGCTCCTCATCGC	
FgDML1-NP-F	GCCAATCTGAATTTCTCGTCGATAAC	For GFP-FgDML1 fusion construct, native promoter
FgDML1-NP-R	TAAATCATCGTCGCTCCTCGTCAC	
FgDML1-GFP-F	TTCAGTGCCAGTGACGAGGACGACGATGATTAATGGTGAGCAAG GGCGAGGA	
FgDML1-GFP-R	TATCTGCCACCAGCCAGCCAACAGCTCCCTTACTTGACAGCTCG TCCA	For GFP-FgDML1 fusion construct, GFP
FgDML1-GR-F	GAGTGGAATAGTTAATCGAGAACGACAC	For GFP-FgDML1 fusion construct, downstream
FgDML1-GR-R	ATCAAGGCTGAAAATGGTGAGAATGGA	
FgDnm1-UF	GAACCACTCCAGTCCGAGA	
FgDnm1-1R	TATCGTCGTCATCTTGTAAATCCAGCACCTCGCCAATAATCTTG	Primer for amplification of up fragment of FgDnm1-1×Flag fusion constructs
FgDnm1-UF	GAACCACTCCAGTCCGAGA	Primer for amplification of up fragment of FgDnm1-2×Flag fusion constructs
FgDnm1-2R	ATCGTCCTTATAGTCCTTATCGTCGTCATCTTGTAAT	
FgDnm1-UF	GAACCACTCCAGTCCGAGA	
FgDnm1-3R	TCATTATCGTCATCATCTTGTAGTCCTTGTATCATCATCGTCTTATA	Primer for amplification of up fragment of FgDnm1-3×Flag fusion constructs
FgDnm1-DF	CGCAAACCGCTCTCCCCAGCAAGCAATGATTATCTCTTGG	Primers for amplification of down fragment of FgDnm1-3×Flag fusion constructs
FgDnm1-DR	GTGATGCCAGAGGAAGACTGATCA	

Fungicides	Concentrations (µg/mL)
Cyazofamid	6.25,12.5,25,50
Pyraclostrobin	0.0016,0.008,0.04,0.2,1,5
Pydiflumetofen	0.00078125,0.0015625,0.003125,0.00625,0.0125
Fluopyram	0.02,0.04,0.08, 0.16, 0.32
Fluxapyroxad	0.02,0.04,0.08,0.16,0.32
Trifloxystrobin	0.0002,0.002,0.02,0.2,2,20

Table. S2. Concentrations of fungicides used in a cross-resistance assay

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MOST National Natural Science Foundation of China (NSFC)	32272585	Yiping Hou

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Additional files

Source data [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

In their study the authors investigated the F. graminearum homologue of the Drosophila Misato-Like Protein DML1 for a function in secondary metabolism and sensitivity to fungicides.

Strengths:

Generally, the topic of the study is interesting and timely and the manuscript is well written, albeit in some cases details on methods or controls are missing.

Weaknesses:

However, a major problem I see is with the core result of the study, the decrease of the DON content associated with deletion of FgDML1: Although some growth data are shown in figure 6 - indicating a severe growth defect - the DON production presented in figure 3 is not related to biomass. Also, the method and conditions for measuring DON are not described. Consequently, it could well be concluded that the decreased amount of DON detected is simply due to a decreased growth and specific DON production of the mutant remains more or less the same.

To alleviate this concern, it is crucial to show the details on the DON measurement and growth conditions and to relate the biomass formation on the same conditions to the DON amount detected. Only then a conclusion as to an altered production in the mutant strains can be drawn.

Comments to the revised manuscript:

The authors carefully revised the manuscript and provided explanations for methods in several cases. However, there are still some problems - probably due to misunderstanding - that need revision.

(1) A major problem of the first version of the manuscript was the lack of appropriate description of biomass analysis and the consideration of the respective results for evaluation of production of DON and other metabolites. Although the authors provide some explanation in the response to reviews, I could not find a corresponding explanation or description in the manuscript. It is not sufficient to explain the problem to me, but a detailed explanation and description of the method has to be provided in the manuscript along with the definition of one "unit of mycelium". It is still not entirely clear to me what such a "unit of mycelium" is.

Please clarify this and any other uncertainties that were commented on by me and other reviewers in the manuscript, not only in the response to reviews. Also adjust the reference list accordingly.

(2) Another problem was, that the authors considered FgDML1 a regulator of DON production. As mentioned by me and reviewer 3, FgDML1 is crucial to numerous functions in *F. graminearum* and its lack causes a plethora of problems for fungal physiology. Hence, although it is clear that the lack of FgDML1 causes alterations in DON production, it is not appropriate to designate this factor as a "regulator".

It seems to me that the authors are afraid that if FgDML1 would not be a "regulator" that this would decrease the value of their study, which is not the case. This is a matter of correct wording. Therefore, please revise the wording accordingly, starting with the title:

...FgDML1 impacts DON toxin biosynthesis...

Moreover, for sure the manuscript might benefit from more detailed description of the whole cascade leading from FgDML1 to DON biosynthesis and production of the other metabolites that change upon deletion. Such explanation can help the reader grasp the relevance of FgDML for regulatory processes as well as on more general versus specific effects.

<https://doi.org/10.7554/eLife.107417.2.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript entitled "Mitochondrial Protein FgDML1 Regulates DON Toxin Biosynthesis and Cyazofamid Sensitivity in *Fusarium graminearum* by affecting mitochondrial homeostasis" identified the regulatory effect of FgDML1 in DON toxin biosynthesis and sensitivity of *Fusarium graminearum* to cyazofamid. The manuscript provides a theoretical framework for understanding the regulatory mechanisms of DON toxin biosynthesis in *F. graminearum* and identifies potential molecular targets for *Fusarium* head blight control. The paper is innovative, but there are issues in the writing that need to be added and corrected.

Comments on revisions:

The author has addressed my questions.

<https://doi.org/10.7554/eLife.107417.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1:

Summary:

*In their study, the authors investigated the *F. graminearum* homologue of the *Drosophila* Misato-Like Protein DML1 for a function in secondary metabolism and sensitivity to fungicides.*

Strengths:

Generally, the topic of the study is interesting and timely, and the manuscript is well written, albeit in some cases, details on methods or controls are missing.

Weaknesses:

However, a major problem I see is with the core result of the study, the decrease in the DON content associated with the deletion of FgDML1. Although some growth data are shown in Figure 6, indicating a severe growth defect, the DON production presented in Figure 3 is not related to biomass. Also, the method and conditions for measuring DON are not described. Consequently, it could well be concluded that the decreased amount of DON detected is simply due to decreased growth, and the specific DON production of the mutant remains more or less the same.

To alleviate this concern, it is crucial to show the details on the DON measurement and growth conditions and to relate the biomass formation under the same conditions to the DON amount detected. Only then can a conclusion as to an altered production in the mutant strains be drawn.

We appreciate it very much that you spent much time on my paper and give me good suggestions, we tried our best to revise the manuscript. I have revised my manuscript according to your suggestions. The point to point responds to the reviewer's comments are listed as following. Our method for DON quantification was based on the amount per unit of mycelium. After obtaining the absorbance value from the ELISA reaction, the concentration of DON was calculated according to a standard curve and a formula, then divided by the dry weight of the mycelium to obtain the DON content per unit of mycelium, with the results finally expressed in µg/g.

(1) Line 139f

... FgDML1 is a critical positive regulator of virulence

Clearly, the deletion of FgDML1 impacts virulence, but it is too much of a general effect to say it is a regulator. DML1 acts high up in the cascade, impacting numerous processes, one of which is virulence. Generally, it has to be considered that deletion of DML1 causes a severe growth defect, which in turn is likely to lead to a plethora of effects. Besides discussing this fact, please also revise the manuscript to avoid references to "direct effects" or "regulator".

Thank you very much for your advice. Our method for determining the amount of DON is based on the amount of mycelium per unit. After obtaining the absorbance value through Elisa reaction, we calculate the concentration of DON toxin according to the established standard curve and formula. Then, we divide it by the dry weight of mycelium to obtain the DON toxin content per unit mycelium, and finally present the results in $\mu\text{g/g}$. In summary, we conclude that the decrease in DON production by ΔFgDML is not due to slower hyphal growth, but rather a decrease in the ability of unit hyphae to produce DON toxins compared to the wild type. Given the decrease in DON toxin synthesis caused by FgDML1 deficiency, we believe that using a regulator is reasonable.

(2) Line 143

Please define "toxin-producing conditions".

Thank you very much for your advice. We have accurately defined the conditions for toxin-producing conditions in the manuscript 'toxin-inducing conditions '(28°C, 145 $\times$ g, 7 days incubation)' (in L163-164)

(3) Line 149

A brief intro on toxisomes should be provided in the introduction to better integrate this into the manuscript's results.

Thank you very much for your advice. We have added corresponding content about toxin producing bodies in the introduction section 'The biosynthesis of DON entails a reorganization of the endoplasmic reticulum into a specialized compartment termed the "toxisome" (Tang et al., 2018). The assembly of the toxisome coincides with the aggregation of key biosynthetic enzymes, which in turn enhances the efficiency of DON production. Concurrently, this compartmentalization serves as a self-defense mechanism, protecting the fungus from the autotoxicity of TRI pathway intermediates (Boenisch et al., 2017). The proteins TRI1, TRI4, TRI14, and Hmr1 are confirmed constituents of this structure(Kistler and Broz, 2015; Menke et al., 2013).' (in L86-93)

(4) Line 153

DON production decreases by about 80 %, but not to 0. Consequently, DML1 is important, but NOT essential for DON production.

Thank you very much for your advice. We have made changes to the wording of the corresponding sections based on your suggestions. 'FgDML1 is essential for the biosynthesis of the DON toxin. '(in L161)

(5) Line 168ff

Please provide a reference for FgDnm1 being critical for mitochondrial fission and state whether such an interaction has been shown in other organisms.

Thank you very much for your advice. We have made changes to the wording of the corresponding sections based on your suggestions. 'FgDnm1 is a key dynamin-related protein mediating mitochondrial fission(Griffin et al., 2005; Kang et al., 2023), suggesting that FgDML1 may form a complex with FgDnm1 to regulate mitochondrial fission and fusion processes. To our knowledge, this is the first report documenting an interaction between DML1 and Dnm in any fungal species, including model organisms such as *S. cerevisiae*. This novel finding provides new insights into the molecular mechanisms underlying mitochondrial dynamics in filamentous fungi. '(in L277-283)

(6) Line 178

Please specify whether Complex III activity was related to biomass and provide a p-value or standard deviation for the value.

Thank you very much for your question. The activity determination of complex III was completed using a complex III enzyme activity kit (Solarbio, Beijing, China) (Li, et al 2022; Wang, et al 2022). Take 0.1 g of standardized mycelium as the sample for the experiment. Given that the mycelium has been homogenized, we believe that there is no necessary correlation between the activity and biomass of complex III. And we also refined the specific measurement steps in the article. ' Briefly, 0.1 g of mycelia was homogenized with 1 mL of extraction buffer in an ice bath. The homogenate was centrifuged at 600 ×g for 10 min at 4°C. The resulting supernatant was then subjected to a second centrifugation at 11,100 ×g for 10 min at 4°C. The pellet was resuspended in 200 µL of extraction buffer and disrupted by ultrasonication (200 W, 5 s pulses with 10 s intervals, 15 cycles). Complex III enzyme activity was finally measured by adding the working solution as per the manufacturer's protocol. Each treatment group contains three biological replicates and three technical replicates. '(in L511-517)

Li C, et al. Amino acid catabolism regulates hematopoietic stem cell proteostasis via a GCN2-eIF2 axis. Cell Stem Cell. 2022 Jul 7; 29(7):1119-1134.e7. doi: 10.1016/j.stem.2022.06.004. PMID: 35803229.

Wang K, et al. Locally organised and activated Fth1hi neutrophils aggravate inflammation of acute lung injury in an IL-10-dependent manner. Nat Commun. 2022 Dec 13;13(1):7703. doi: 10.1038/s41467-022-35492-y. PMID: 36513690; PMCID: PMC9745290

(7) Line 185

Albeit this headline is a reasonable hypothesis, you actually did not show that the conformation is altered. Please reword accordingly.

Please also add references for cyazofamid acting on the QI site versus other fungicides acting on the QO site.

Thank you very much for your advice. We have made changes to the wording of the corresponding sections based on your suggestions. 'Overexpression of *FgQCR2*, *FgQCR8*, and *FgQCR9* may alters the conformation of the QI site, resulting in reduced sensitivity to cyazofamid. '(in L212-213). For fungicides targeting Qi and QO sites, we have added corresponding descriptions in the respective sections 'Numerous fungicides have been developed to inhibit the Qo site (e.g., pyraclostrobin, azoxystrobin)(Nuwamanya et al., 2022; Peng et al., 2022) and the Qi site (e.g., cyazofamid)(Mitani et al., 2001) of the cytochrome bc1 complex. '(in L327-329)

(8) Line 200

This section on growth should be moved up right after introducing the mutant strain.

Thank you very much for your advice. We have advanced the part of nutritional growth and sexual asexual development before DON toxin to promote better reading and understanding. We arranged the sequence in the previous way to emphasize the new discovery between mitochondria and DON toxin. We found a significant decrease in DON toxin in ΔFgDML1, defects in the formation of toxin producing bodies, and downregulation of FgTRIs at both the gene and protein levels. In summary, we believe that the absence of FgDML1 does indeed lead to a decrease in the content of DON toxin, and FgDML1 plays a regulatory role in the synthesis of DON toxin. In addition, our measurements of DON toxin, acetyl CoA, ATP and other indicators are all based on the amount per unit hyphae, excluding differences caused

by hyphal biomass or growth. We have further refined the materials and methods to facilitate better reading and understanding.

(9) Line 203

"... significantly reduced growth rates ..."

This is not what was measured here. Figure 6A shows a plate assay that can be used to assess hyphal extension. In the figure, it is also visible that the mycelium of the deletion mutant is much denser, maybe due to increased hyphal branching. Please reword.

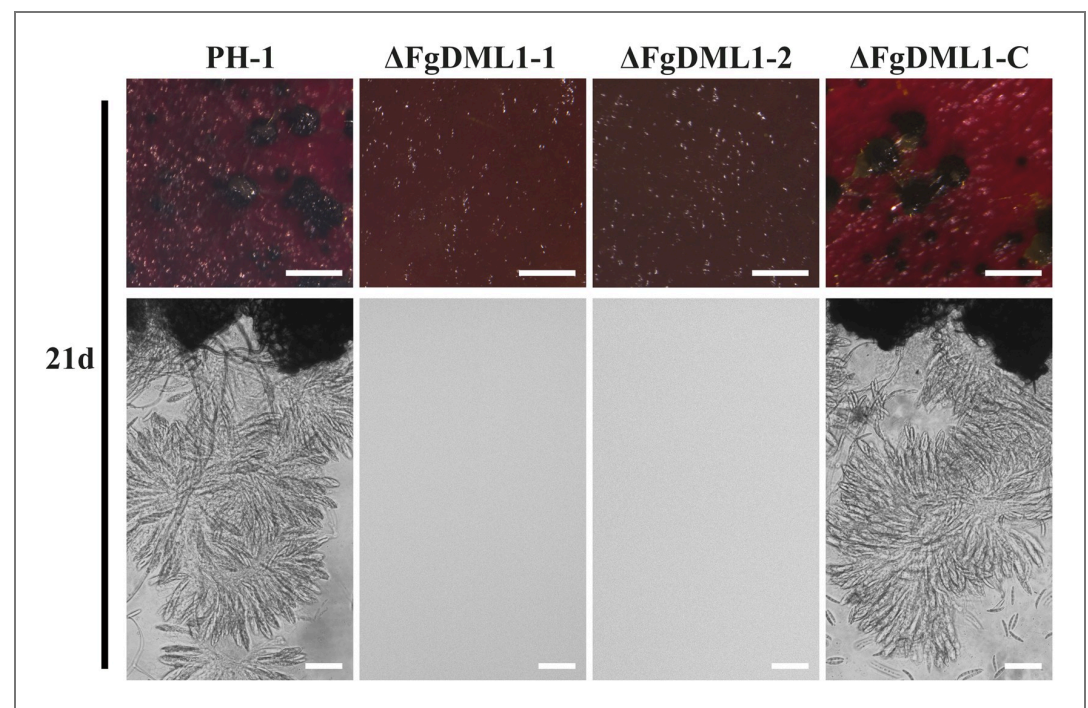
Additionally, it is important to include a biomass measurement here under the conditions used for DON assessment. Hyphal extension measurements cannot be used instead of biomass.

Thank you very much for your advice. We have made changes to the wording of the corresponding sections based on your suggestions. 'The Δ FgDML1 strain displayed a distinct growth phenotype characterized by retardation in radial growth and the formation of more compact, denser hyphal networks on all tested media compared to the PH-1 and Δ FgDML-C strains.' (in L136-138).

(10) Line 217

Please include information on how long the cultures were monitored. Given the very slow growth of the mutant, perithecia formation may be considerably delayed beyond 14 days.

Thank you very much for your advice. Based on your suggestion, we have extended the incubation time for sexual reproduction to 21 days to more accurately evaluate its sexual reproduction ability. Our results show that even after 21 days, Δ FgDML1 still cannot produce ascospores and ascospores, which proves that the absence of FgDML1 does indeed cause sexual reproduction defects in *F. graminearum*.



Author response image 1.

Discussion

(11) Please mention your summary Figure 8 early on in the discussion, and explain conclusions with this figure in mind. Please avoid repetition of the results section as much as possible.

Also, please state clearly what was already known from previous research and is in agreement with your results, and what is new (in fungi or generally).

Thank you very much for your advice. Based on your suggestion, we mentioned Fig8 earlier in the first half of the discussion and provided guidance for the following text. We also conducted a more comprehensive discussion by analyzing our research results and comparing them with previous studies. 'Our study defines a novel mechanism through which FgDML1 governs mitochondrial homeostasis. We demonstrate that FgDML1 directly interacts with the key mitochondrial fission regulator FgDnm1 and positively modulates cellular bioenergetic metabolism, as evidenced by elevated ATP and acetyl-CoA levels (Fig. 8). '(in L250-253). 'The Misato/DML1 protein family is evolutionarily conserved from yeast to humans and plays a critical role in mitochondrial regulation. In *S. cerevisiae*, DML1 is an essential gene; its deletion is lethal, while its overexpression results in fragmented mitochondrial networks and aberrant cellular morphology, underscoring its necessity for normal mitochondrial function (Gurvitz et al., 2002). Similarly, in *Homo sapiens*, the homolog Misato localizes to the mitochondrial outer membrane, and both its depletion and overexpression are sufficient to disrupt mitochondrial morphology and distribution (Kimura and Okano, 2007). '(in L241-244).

(12) Line 262ff

Please specify if this interaction was shown previously in other organisms and provide references.

Thank you very much for your advice. We have clearly stated in the corresponding section that the interaction between FgDML and FgDnm is the first reported, and to our knowledge, no relevant reports have been found in other species so far. ' Notably, FgDML1 was found to interact with FgDnm1 (Fig. 5E), FgDnm1 is a key dynamin-related protein mediating mitochondrial fission(Griffin et al., 2005; Kang et al., 2023), suggesting that FgDML1 may form a complex with FgDnm1 to regulate mitochondrial fission and fusion processes. To our knowledge, this is the first report documenting an interaction between DML1 and Dnm in any fungal species, including model organisms such as *S. cerevisiae*. This novel finding provides new insights into the molecular mechanisms underlying mitochondrial dynamics in filamentous fungi. '(in L276-283)

(13) Line 287ff

There is no result that would justify this speculation. Please remove.

Thank you very much for your advice. We have modified the corresponding wording in the corresponding section. 'In conclusion, our findings suggest that the overexpression of assembly factors *FgQCR2*, *FgQCR7*, and *FgQCR8* in Δ FgDML1 potentially modifies the conformation of the Qi site, which specifically modulates the sensitivity of *F. graminearum* to cyazofamid. '(in L352-355)

Materials and methods

(14) A table with all primer sequences used in the study and their purpose is missing. For every experiment, the number of technical and biological replicates needs to be stated.

Thank you very much for your advice. We have presented all the primers used in this study in Supplementary Table 1 (in Table S1). We added the number of technical and biological replicates in the material and method descriptions for each experiment. 'For each sample, a total of 200 conidia were counted. The experiment included three biological replicates with three technical replicates each.' (in L434-436). 'Each treatment group contains three biological replicates.' (in L444-445). 'Each treatment group contains three biological replicates and three technical replicates.' (in L463-464). 'Each treatment group contains three biological replicates and three technical replicates.' (in L474-475). 'Each treatment group contains three biological replicates.' (in L483). 'Each treatment group contains three biological replicates and three technical replicates.' (in L501-502). 'Each treatment group contains three biological replicates and three technical replicates.' (in L516-517). 'The experiment was independently repeated three times.' (in L533-534).

(15) Line 369ff

Please provide final concentrations used for assays here.

Thank you very much for your advice. The final concentration has been displayed in the Figure (in Fig6. A, B) (in Fig. S3). And we have provided supplementary Table 2 to reflect the concentration in a more intuitive way. (in Table. S2)

(16) Line 383

Please provide a reference or data on the use of F2du for transformant selection and explain the abbreviation.

Thank you very much for your advice. Based on your suggestion, we have provided the full name and references of F2du. 'Transformants were selected on PDA plates containing either 100 µg/mL Hygromycin B (Yeasen, Shanghai, China) or 0.2 µmol/mL 5-Fluorouracil 2'-deoxyriboside (F2du) (Solarbio, Beijing, China) (Zhao et al., 2022).' (in L405-407).

(17) Line 407

Please provide a reference for the method and at least a brief description.

Thank you very much for your advice. Based on your suggestion, we have added references and provided a brief introduction to the method. 'As previously described (Tang et al., 2020; Wang et al., 2025), Specifically, coleoptiles were inoculated with conidial suspensions and incubated for 14 days, while leaves were inoculated with fresh mycelial plugs and incubated for 5 days, followed by observation and quantification of disease symptoms. DON toxin was measured using a Wise Science ELISA-based kit (Wise Science, Jiangsu, China) (Li et al., 2019; Zheng et al., 2018).' (in L466-471)

(18) Line 414ff

Also, here, the amount of biomass has to be considered for the measurement to be able to distinguish if actually less of the compounds were produced or if the effect seen was merely due to an altered amount of biomass present.

Thank you very much for your advice. We believe that biomass is not within the scope of our measurement indicators, as we have measured and calculated based on unit hyphae. Therefore, we have ruled out experimental bias caused by a decrease in biomass.

RNA and RT-qPCR

(19) Line 461

When the strains were transferred to AEA medium, was the biomass measured, at least wet weight, and in which culture volume was it done? It makes a big difference if the amount of (wet) biomass dilutes a small amount of fungicide-containing culture or if biomass is added in at least roughly equal amounts in sufficient growth medium to ensure equal conditions.

Thank you very much for your question. Our sample processing controlled the wet weight of the samples before dosing, ensuring that the wet weight of the mycelium obtained from each sample before dosing was 0.2g. The mycelium was obtained through AEA with a volume of 100mL. This ensured consistency in the initial biomass between groups before dosing, and also ensured the accuracy of the drug concentration.

(20) Line 466

Please provide the name and supplier of the kit.

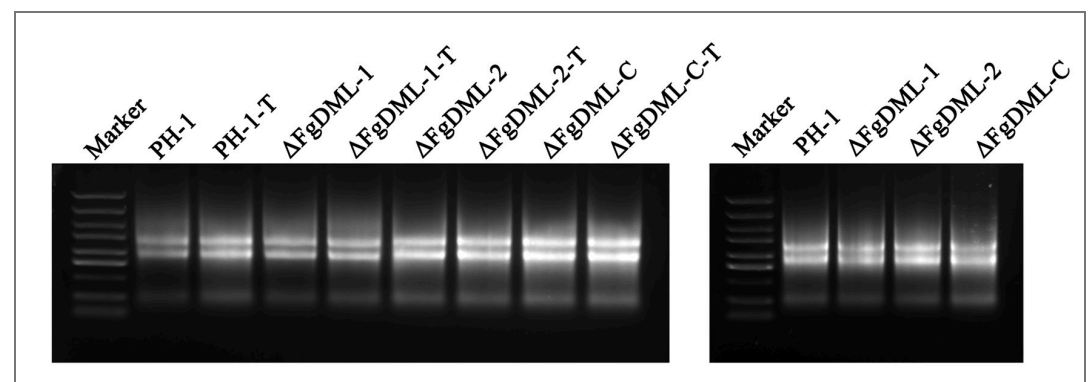
Thank you very much for your advice. We have added corresponding content in the corresponding location. 'Mycelium was collected and total RNA was extracted following the instructions provided by the Total RNA Extraction Kit (Tiangen, Beijing, China).' (in L523-524).

(21) All primer sequences must be provided in a table.

Thank you very much for your advice. We have presented all the primers used in this study in Supplementary Table 1. (in Table S1).

(22) For RT qPCR it is essential to check the RNA quality to be sure that the obtained results are not artifacts due to varying quality, which may exceed differences. Please state how quality control was done and which threshold was applied for high-quality RNA to be used in RTqPCR (like RIN factor, etc).

Thank you very much for your question. We performed stringent quality control on the extracted total RNA. First, a micro-spectrophotometer was used to measure RNA concentration and purity, confirming that the A260/A280 ratio was between 1.8 and 2.0 and the A260/A230 ratio was greater than 2.0, indicating good RNA purity without significant protein or organic solvent contamination. Subsequently, verification by agarose gel electrophoresis revealed distinct 28S and 18S rRNA bands, demonstrating good RNA integrity and absence of degradation.



Author response image 2.

(B): Minor Comments:

(1) Please increase the font size of the labels and annotations of the figures; it is hard to read as it is now.

Thank you very much for your advice. We have increased the size of annotations or numerical labels in the corresponding images for better reading.

(2) Throughout the manuscript: Please check that all abbreviations are explained at first use.

Thank you very much for your advice. We have checked the entire text to ensure that abbreviations have their full names when they first appear.

(3) I do hope that the authors can clarify all concerns and provide an amended manuscript of this interesting story.

Thank you very much for your advice. Sincerely thank you for your suggestions and questions, which have been very helpful to us.

Reviewer #2:

The manuscript entitled "Mitochondrial Protein FgDML1 Regulates DON Toxin Biosynthesis and Cyazofamid Sensitivity in Fusarium graminearum by affecting mitochondrial homeostasis" identified the regulatory effect of FgDML1 in DON toxin biosynthesis and sensitivity of Fusarium graminearum to cyazofamid. The manuscript provides a theoretical framework for understanding the regulatory mechanisms of DON toxin biosynthesis in F. graminearum and identifies potential molecular targets for Fusarium head blight control. The paper is innovative, but there are issues in the writing that need to be addressed and corrected.

We appreciate it very much that you spent much time on my paper and give me good suggestions, we tried our best to revise the manuscript. I have revised my manuscript according to your suggestions with red words. In the response comments, to highlight the specific positions of the revised parts in the manuscript with red line number. The point to point responds to the reviewer's comments are listed as following.

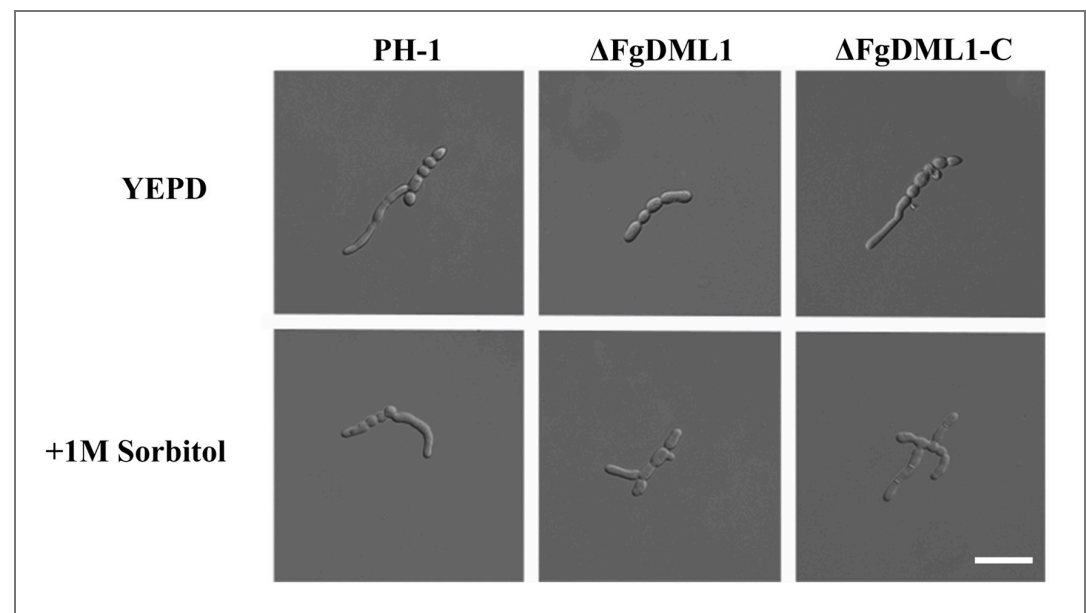
Weaknesses:

(1) *The authors speculate that cyazofamid treatment caused upregulation of the assembly factors, leading to a change in the conformation of the Qi protein, thus restoring the enzyme activity of complex III. But no speculation was given in the discussion as to why this would lead to the upregulation of assembly factors, and how the upregulation of assembly factors would change the protein conformation, and is there any literature reporting a similar phenomenon? I would suggest adding this to the discussion.*

Thank you very much for your advice. Based on your suggestion, we have added content related to the assembly factor of complex III in the discussion section and made modifications to the corresponding wording. 'Previous studies have reported that mutations in the Complex III assembly factors TTC19, UQCC2, and UQCC3 impair the assembly and activity of Complex III (Feichtinger et al., 2017; Wanschers et al., 2014). '(in L345-347). 'In conclusion, our findings suggest that the overexpression of assembly factors *FgQCR2*, *FgQCR7*, and *FgQCR8* in Δ FgDML1 potentially modifies the conformation of the Qi site, which specifically modulates the sensitivity of *F. graminearum* to cyazofamid. '(in L352-355).

(2) *Would increased sensitivity of the mutant to cell wall stress be responsible for the excessive curvature of the mycelium?*

Thank you very much for your question. We believe that the sensitivity of Δ FgDML1 to osmotic stress is reduced, which may not be related to hyphal bending, as shown in the Author response image 3. During the conidia stage, Δ FgDML1 cannot germinate in YEPD, while the application of 1M Sorbitol promotes its germination. But it is caused by internal unknown mechanisms, which is also the focus of our future research.



Author response image 3.

(3) The vertical coordinates of Figure 7B need to be modified with positive inhibition rates for the mutants.

Thank you very much for your advice. The display in Figure 7B truly reflects its inhibition rate. In the Δ FgDML1 mutant, when subjected to osmotic stress treatment, the inhibition rate becomes negative, indicating that the colony growth is greater than that of the CK. Therefore, the negative inhibition rate is shown in Figure 7B.

(1) In Figure 1B, Figure 3C, and Figure 6C, the scale below the picture is not clear. In Figure 5D, the histogram is unclear, and it is recommended to redraw the graph.

Thank you very much for your advice. The issue with the above images may be due to Word compression. We have changed the settings and enlarged the images as much as possible to better display them.

(2) The full Latin name of the strain should be used in the title of figures and tables.

Thank you very much for your advice. Based on your suggestion, we have used the full names of the strains appearing in the title of figures and tables.

(3) Proteins in line 117 should be abbreviated.

Thank you very much for your advice. Based on your suggestion, we have abbreviated the corresponding positions. "The DML1 protein from *S. cerevisiae* was used as a query for a BLAST search against the *Fusarium* genome database, resulting in the identification of the putative DML1 gene FgDML1 (FGSG_05390) in *F. graminearum*." (in L118-120).

(4) The sentence in lines 187-189, which is supposed to introduce why the test is sensitive to the three drugs, is currently illogical.

Thank you very much for your advice. Based on your suggestion, we have made modifications to the corresponding sections. 'Since Complex III is involved in the action of both cyazofamid (targeting the QI site) and pyraclostrobin (targeting the QO site), the sensitivity of Δ FgDML1 to cyazofamid and pyraclostrobin was investigated. ' (in L214-216).

(5) The expression of FgQCR2, FgQCR7, and FgQCR8 was significantly upregulated in Δ FgDML1 at transcription levels. Do FgQCR2, FgQCR8, and FgQCR9 show upregulated expression at the protein level?

Thank you very much for your question. Based on your suggestion, we evaluated the protein expression levels of FgQCR2, FgQCR7, and FgQCR8 in PH-1 and Δ FgDML1, and we found that the protein expression levels of FgQCR2, FgQCR7, and FgQCR8 in Δ FgDML1 were higher than those in PH-1. (in Fig. 6F).

(6) In Figure 7B, it is recommended to adjust the position of the horizontal axis labels in the histogram.

Thank you very much for your advice. Based on your suggestion, we have made modifications to the corresponding sections.(in Fig. 7B)

(7) There are numerous errors in the writing of gene names in the text. Please check the full text and change the writing of gene names and mutant names to italic.

Thank you very much for your advice. We have checked the entire text to ensure that all genes have been italicized.

(8) All acronyms should be spelled out in figure and table captions. e.g., *F. graminearum*.

Thank you very much for your advice. Based on your suggestion, we have used the full names of the strains appearing in the title of figures and tables.

(9) In line 492, *P* should be lowercase and italic.

Thank you very much for your advice. Based on your suggestion, we have made adjustments to the corresponding content.

Reviewer #3:

Summary:

The manuscript "Mitochondrial 1 protein FgDML1 regulates DON toxin biosynthesis and cyazofamid sensitivity in Fusarium graminearum by affecting mitochondrial homeostasis" describes the construction of a null mutant for the FgDML1 gene in F. graminearum and assays characterising the effects of this mutation on the pathogen's infection process and lifecycle. While FgDML1 remains underexplored with an unclear role in the biology of filamentous fungi, and although the authors performed several experiments, there are fundamental issues with the experimental design and execution, and interpretation of the results.

Strengths:

FgDML1 is an interesting target, and there are novel aspects in this manuscript. Studies in other organisms have shown that this protein plays important roles in mitochondrial DNA (mtDNA) inheritance, mitochondrial compartmentalisation, chromosome segregation, mitochondrial distribution, mitochondrial fusion, and overall mitochondrial

dynamics. Indeed, in *Saccharomyces cerevisiae*, the mutation is lethal. The authors have carried out multi-faceted experiments to characterise the mutants.

Weaknesses:

However, I have concerns about how the study was conceived. Given the fundamental importance of mitochondrial function in eukaryotic cells and how the absence of this protein impacts these processes, it is unsurprising that deletion of this gene in *F. graminearum* profoundly affects fungal biology. Therefore, it is misleading to claim a direct link between FgDML1 and DON toxin biosynthesis (and virulence), as the observed effects are likely indirect consequences of compromised mitochondrial function. In fact, it is reasonable to assume that the production of all secondary metabolites is affected to some extent in the mutant strains and that such a strain would not be competitive at all under non-laboratory conditions. The order in which the authors present the results can be misleading, too. The results on vegetative growth rate appeared much later in the manuscript, which should have come first, as the FgDML1 mutant exhibited significant growth defects, and subsequent results should be discussed in that context. Moreover, the methodologies are not described properly, making the manuscript hard to follow and difficult to replicate.

We appreciate it very much that you spent much time on my paper and give me good suggestions, we tried our best to revise the manuscript. I have revised my manuscript according to your suggestions with red words. In the response comments, to highlight the specific positions of the revised parts in the manuscript with red line number. The point to point responds to the reviewer's comments are listed as following.

For weaknesses, we arranged the sequence in this way to emphasize the novel discovery between mitochondria and DON toxin. We found a significant decrease in DON toxin in Δ FgDML1, defects in the formation of toxin producing bodies, and downregulation of FgTRIs at both the gene and protein levels. In summary, we believe that the absence of FgDML1 does indeed lead to a decrease in the content of DON toxin, and FgDML1 plays a regulatory role in the synthesis of DON toxin. In addition, our measurements of DON toxin, acetyl CoA, ATP and other indicators are all based on the amount per unit hyphae, excluding differences caused by hyphal biomass or growth. We have further refined the materials and methods to facilitate better reading and understanding.

(1) Lines 37-39: *The disease itself does not produce toxins; it is the fungus that causes the disease that produces toxins. Moreover, the disease symptoms observed are likely caused by the toxins produced by the fungus.*

Thank you very much for your advice. We have made modifications to the wording of the corresponding sections. 'Studies have shown that increased DON levels are positively correlated with the pathogenicity rate of *F. graminearum*.' (in L36-37).

(2) Lines 82-87: *While it is challenging to summarise the role of ATP in just a few words, this section needs improvement for clarity and accuracy. Additionally, I do not believe that drawing a direct link between mitochondrial defects and toxin production is an appropriate strategy in this case.*

Thank you very much for your advice. Based on your suggestion, we have added corresponding descriptions in the corresponding positions to provide more information on the relationship between ATP and toxins, in order to better prepare for the following text. 'Pathogen-intrinsic ATP homeostasis is recognized as a critical, rate-limiting determinant for toxin biosynthesis. Previous studies indicate that dual-target inhibition of ATP synthase (AtpA) and adenine deaminase (Ade) by a specific small-molecule probe effectively depletes intracellular ATP, consequently suppressing the synthesis of key virulence factors TcdA and TcdB transcriptionally and translationally (Marreddy et al., 2024). The systemic toxicity of

Anthrax Edema Toxin (ET) is primarily attributed to its catalytic activity, which depletes the host cell's ATP reservoir, thereby triggering a bioenergetic collapse that culminates in cell lysis and death (Liu et al., 2025). '(in L78-86).

(3) Lines 125-126: The manuscript does not clearly describe how subcellular localisation was determined. This methodology needs to be properly detailed.

Thank you very much for your advice. The subcellular localization was validated through co-localization analysis with MitoTracker Red CMXRos, a mitochondrial-specific dye. The observed overlap between the FgDML1-GFP signal and the mitochondrial marker confirmed mitochondrial localization. Based on these results, we determined that FgDML1 is definitively localized to the mitochondria. We have incorporated this description in the appropriate section of the manuscript. 'Furthermore, subcellular localization studies confirmed that FgDML1 localizes to mitochondria, as demonstrated by colocalization with a mitochondria-specific dye MitoTracker Red CMXRos (Fig. 1B). '(in L125-127).

(4) Regarding the organisation of the Results section, it needs to be revised. While I understand the authors' intention to emphasise the impact on virulence, the results showing how FgDML1 deletion affects vegetative growth, asexual and sexual reproduction, and sensitivity to stressors should be presented before the virulence assays and effects on DON production. Additionally, the authors do not provide any clear evidence that FgDML1 directly interacts with proteins involved in asexual or sexual reproduction, stress responses, or virulence. Therefore, it is misleading to suggest that FgDML1 directly regulates these processes. The observed phenotypes are, rather, a consequence of severely impaired mitochondrial function. Without functional mitochondria, the cell cannot operate properly, leading to widespread physiological defects. In this regard, statements such as those in lines 139-140 and 343-344 are misleading.

Thank you very much for your advice. We have adjusted the order of the images based on your suggestion, placing the characterization of Δ FgDML1 in nutritional growth, sexual reproduction, and other aspects before DON toxin. And we have made adjustments to the corresponding statements. 'These findings demonstrate that FgDML1 is a positive regulator of virulence in *F. graminearum*. '(in L140-141).

(5) Lines 185-186: The authors do not provide sufficient evidence to support the claim that FgQCR2, FgQCR8, and FgQCR9 overexpression is the main cause of reduced cyazofamid sensitivity. Although expression of these genes is altered, reduced sensitivity may result from changes in other proteins or pathways. To strengthen this claim, overexpression of FgQCR2, 8, and 9 in the wild-type background, followed by assessment of cyazofamid resistance, would be necessary. As it stands, there is no support for the claim presented in lines 329-332.

Thank you very much for your advice. To establish a causal link between the overexpression of FgQCR2, FgQCR7, and FgQCR8 and the observed reduction in cyazofamid sensitivity, we first quantified the protein levels of these assembly factor. Western blot analysis confirmed their elevated expression in the Δ FgDML1 mutant compared to the wild-type PH-1. We further generated individual overexpression strains for FgQCR2, FgQCR7, and FgQCR8 in the wild-type PH-1 background. Fungicide sensitivity assays revealed that all three overexpression mutants displayed significantly reduced sensitivity to cyazofamid compared to the parental strain. These genetic complementation experiments confirm that upregulation of FgQCR2, FgQCR7, and FgQCR8 is sufficient to confer reduced cyazofamid sensitivity. We have incorporated these explanations and provided supporting images in the appropriate section of the manuscript. 'To further clarify whether the upregulated expression of *FgQCR2*, *FgQCR7*, and *FgQCR8* genes affects their protein expression levels, we measured the protein levels. The results showed that the protein expression levels of FgQCR2, FgQCR7, and FgQCR8 in

Δ FgDML1 were higher than those in PH-1(Fig. 6F). Subsequently, we overexpressed *FgQCR2*, *FgQCR7*, and *FgQCR8* in the wild-type background, and the corresponding overexpression mutants exhibited reduced sensitivity to cyazofamid(Fig. 6E). '(in L205-211)(in Fig. 6E, F)

(6) Lines 187-190: This segment is confusing and difficult to follow. It requires rewriting for clarity.

Thank you very much for your advice. Based on your suggestion, we have made corresponding modifications in the corresponding locations. 'Since Complex III is involved in the action of both cyazofamid (targeting the QI site) and pyraclostrobin (targeting the QO site), the sensitivity of Δ FgDML1 to cyazofamid and pyraclostrobin was investigated. '(in L214-216)

(7) Lines 345-346: The authors state that in this study, *FgDML1* is localised in mitochondria, which implies that in other studies, its localisation was different. Is this accurate? Clarification is needed.

Thank you very much for your question. In previous studies, the localization of this protein was not clearly defined, and its function was only emphasized to be related to mitochondria. Whether in yeast or in *Drosophila melanogaster*. (Miklos et al., 1997; Gurvitz et al., 2002)

Miklos GLG, Yamamoto M-T, Burns RG, Maleszka R. 1997. An essential cell division gene of *drosophila*, absent from *saccharomyces*, encodes an unusual protein with tubulin-like and myosin-like peptide motifs. *Proc Natl Acad Sci* 94:5189–5194. doi:10.1073/pnas.94.10.5189

Gurvitz A, Hartig A, Ruis H, Hamilton B, de Couet HG. 2002. Preliminary characterisation of DML1, an essential *saccharomyces cerevisiae* gene related to *misato* of *drosophila melanogaster*. *FEMS Yeast Res* 2:123–135. doi:10.1016/S1567-1356(02)00083-1

Material and Methods Section

(8) In general, the methods require more detailed descriptions, including the brands and catalog numbers of reagents and kits used. Simply stating that procedures were performed according to manufacturers' instructions is insufficient, particularly when the specific brand or kit is not identified.

Thank you very much for your advice. We have added corresponding content based on your suggestion to more comprehensively display the reagent brand and complete product name. 'Transformants were selected on PDA plates containing either 100 μ g/mL Hygromycin B (Yeasen, Shanghai, China) or 0.2 μ mol/mL 5-Fluorouracil 2'-deoxyriboside (F2du) (Solarbio, Beijing, China)(Zhao et al., 2022). ' (in L405-407). 'DON toxin was measured using a Wise Science ELISA-based kit (Wise Science, Jiangsu, China) (Li et al., 2019; Zheng et al., 2018) '. (in L469-471)

(9) Line 364: What do CM and MM stand for? Please define.

Thank you very much for your advice. Based on your suggestion, we have made modifications in the corresponding locations. 'To evaluate vegetative growth, complete medium (CM), minimal medium (MM), and V8 Juice Agar (V8) media were prepared as described previously(Tang et al., 2020). '(in L385-387)

Generation of Deletion and Complemented Mutants:

(10) This section lacks detail. For example, were PCR products used directly for PEG-mediated transformation, or were the fragments cloned into a plasmid?

Thank you very much for your question. We directly use the fused fragments for protoplast transformation after sequencing confirmation. We have clearly defined the fragment form

used for transformation at the corresponding location. 'The resulting fusion fragment was transformed into the wild-type *F. graminearum* PH-1 strain via polyethylene glycol (PEG)-mediated protoplast transformation.' (in L403-405).

(11) *PCR and Southern blot validation results should be included as supplementary material, along with clear interpretations of these results.*

Thank you very much for your advice. In the supplementary material we submitted, Supplementary Figure 2 already includes the results of PCR and Southern blot validation. (in Fig. S2)

(12) *There is almost no description of how the mutants mentioned in lines 388-390 were generated.*

Thank you very much for your advice. Based on your suggestions, we have added relevant content in the appropriate sections to more comprehensively and clearly reflect the experimental process. 'Specifically, FgDML1, including its native promoter region and open reading frame (ORF) (excluding the stop codon), was amplified. The PCR product was then fused with the *Xho*I-digested pYF11 vector. After transformation into *E. coli* and sequence verification, the plasmid was extracted and subsequently introduced into PH-1 protoplasts. For FgDnm1-3×Flag, the 3×Flag tag was added to the C-terminus of FgDnm1 by PCR, fused with the hygromycin resistance gene and the FgDnm1 downstream arm, and then introduced into PH-1 protoplasts. The overexpression mutant was constructed according to a previously described method. Specifically, the ORF of FgDML1 was amplified and the PCR product was ligated into the *Sac*II-digested pXS overexpression vector. The resulting plasmid was then transformed into PH-1 protoplasts (Shi et al., 2023). For the construction of PH-1::FgTri1+GFP and ΔFgDML1::FgTri1+GFP, the ORF of FgTri1 was amplified and ligated into the *Xho*I-digested pYF11 vector as described above. The resulting vectors were then transformed into protoplasts of PH-1 or ΔFgDML1, respectively.' (in L413-426).

Vegetative Growth and Conidiation Assays:

(13) *There is no information about how long the plates were incubated before photos were taken. Judging by the images, it appears that different incubation times may have been used.*

Thank you very much for your advice. Due to the slower growth of ΔFgDML1, we adopted different incubation periods and have supplemented the relevant content in the corresponding section. 'All strains were incubated at 25°C in darkness; however, due to ΔFgDML1 slower growth, the ΔFgDML1 mutant required a 5-day incubation period compared to the 3 days used for PH-1 and ΔFgDML1-C.' (in L490-493).

(14) *There is no description of the MBL medium.*

Thank you very much for your advice. Based on your suggestion, we have supplemented the corresponding content in the corresponding positions. 'Mung bean liquid (MBL) medium was used for conidial production, while carrot agar (CA) medium was utilized to assess sexual reproduction (Wang et al., 2011).' (in L387-389).

DON Production and Pathogenicity Assays:

(15) *Were DON levels normalised to mycelial biomass? The vegetative growth assays show that FgDML1 null mutants exhibit reduced growth on all tested media. If mutant and wild-type strains were incubated for the same period under the same conditions, it is reasonable to assume that the mutants accumulated significantly less biomass. Therefore, results related to DON production, as well as acetyl-CoA and ATP levels, must be normalised to biomass.*

Thank you very much for your question. We have taken into account the differences in mycelial biomass. Therefore, when measuring DON, acetyl-CoA, and ATP levels, all data were normalized to mycelial mass and calculated as amounts per unit of mycelium, thereby avoiding discrepancies arising from variations in biomass.

Sensitivity Assays:

(16) While the authors mention that gradient concentrations were used, the specific concentrations and ranges are not provided. Importantly, have the plates shown in Figure 5 been grown for different periods or lengths? Given the significantly reduced growth rate shown in Figure 6A, the mutants should not have grown to the same size as the WT (PH-1) as shown in Figures 5A and 5B unless the pictures have been taken on different days. This needs to be explained.

Thank you very much for your question. Due to the slower growth of Δ FgDML1, we adopted different incubation periods and have supplemented the relevant content in the corresponding section. 'All strains were incubated at 25°C in darkness; however, due to Δ FgDML1 slower growth, the Δ FgDML1 mutant required a 5-day incubation period compared to the 3 days used for PH-1 and Δ FgDML1-C. '(in L490-493).

(17) Additionally, was inhibition measured similarly for both stress agents and fungicides? This should be clarified.

Thank you very much for your question. We have supplemented the specific concentration gradient of fungicides. 'The concentration gradients for each fungicide in the sensitivity assays were set up according to Supplementary Table S2. '(in L493-494)(in Table. S2).

Complex III Enzyme Activity:

(18) A more detailed description of how this assay was performed is needed.

Thank you very much for your advice. We have provided further detailed descriptions of the corresponding sections. 'Briefly, 0.1 g of mycelia was homogenized with 1 mL of extraction buffer in an ice bath. The homogenate was centrifuged at 600 $\times$ g for 10 min at 4°C. The resulting supernatant was then subjected to a second centrifugation at 11,000 $\times$ g for 10 min at 4°C. The pellet was resuspended in 200 μ L of extraction buffer and disrupted by ultrasonication (200 W, 5 s pulses with 10 s intervals, 15 cycles). Complex III enzyme activity was finally measured by adding the working solution as per the manufacturer's protocol. '(in L511-517)

(19) Were protein concentrations standardised prior to the assay?

Thank you very much for your question. Protein concentrations for all Western blot samples were quantified using a BCA assay kit to ensure equal loading.

(20) Line 448: Are Δ FgDML1::Tri1+GFP and Δ FgDML1+GFP the same strain? Δ FgDML1::Tri1+GFP has not been previously described.

Thank you very much for your question. These two strains are not the same strain, and we have supplemented their construction process in the corresponding section. 'For the construction of PH-1::FgTri1+GFP and Δ FgDML1::FgTri1+GFP, the ORF of FgTri1 was amplified and ligated into the *Xho*I-digested pYF11 vector as described above. The resulting vectors were then transformed into protoplasts of PH-1 or Δ FgDML1, respectively. '(in L423-426)

(21) Lines 460 and 468: Please adopt a consistent nomenclature, either RT-qPCR or qRT-PCR.

Thank you very much for your advice. We have unified it and modified the corresponding content in the corresponding sections. 'Reverse Transcription Quantitative Polymerase Chain Reaction (RT-qPCR) was carried out using the QuantStudio 6 Flex real-time PCR system (Thermo, Fisher Scientific, USA) to assess the relative expression of three subunits of Complex III (*FgCytb*, *FgCyt1*, *FgISP*), five assembly factors (*FgQCR2*, *FgQCR6*, *FgQCR7*, *FgQCR8*, *FgQCR9*), and DON biosynthesis-related genes (*FgTri5* and *FgTri6*). '(in L526-531)

(22) Lines 472-473: Why was *FgCox1* used as a reference for *FgCytb*? Clarification is needed.

Thank you very much for your question. *FgCytb* (cytochrome b) and *FgCOX1* (cytochrome c oxidase subunit I) are both encoded by the mitochondrial genome and serve as core components of the oxidative phosphorylation system (Complex III and Complex IV, respectively). Their transcription is co-regulated by mitochondrial-specific mechanisms in response to cellular energy status. Consequently, under experimental conditions that perturb energy homeostasis, *FgCOX1* expression exhibits relative, context-dependent stability with *FgCytb*, or at least co-varies directionally, making it a superior reference for normalizing target gene expression. In contrast, *FgGapdh* operates within a distinct genetic and regulatory system. Using *FgCOX1* ensures that both reference and target genes reside within the same mitochondrial compartment and functional module, thereby preventing normalization artifacts arising from independent variation across disparate pathways.

(23) Lines 476-477: This step requires a clearer and more detailed explanation.

Thank you very much for your advice. We provided detailed descriptions of them in their respective positions. 'For *FgDnm1*-3×Flag, the 3×Flag tag was added to the C-terminus of *FgDnm1* by PCR, fused with the hygromycin resistance gene and the *FgDnm1* downstream arm, and then introduced into PH-1 protoplasts. '(in L417-419). 'The *FgDnm1*-3×Flag fragment was introduced into PH-1 and *FgDML1*+GFP protoplasts, respectively, to obtain single-tagged and double-tagged strains. '(in L541-543)

Western blotting:

(24) Uncropped Western blot images should be provided as supplementary material.

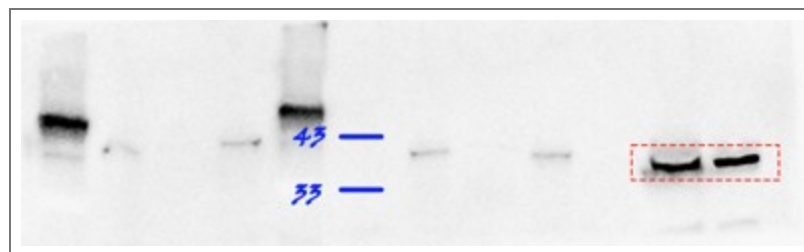
Thank you very much for your advice. All Western blot images will be submitted to the supplementary material package.

(25) Lines 485-489: A more thorough description of the antibodies used (including source, catalogue number, and dilution) is necessary.

Thank you very much for your advice. The antibodies used are clearly stated in terms of brand, catalog number, and dilution. We have added the dilution ratio. 'All antibodies were diluted as follows: primary antibodies at 1:1000 and secondary antibodies at 1:10000. '(in L550-551)

(26) The Western blot shown in Figure 3D appears problematic, particularly the anti-GAPDH band for *FgDML1::FgTri1*+GFP. Are both anti-GAPDH bands derived from the same gel?

Thank you very much for your advice. We are unequivocally certain that these data derive from the same gel. Therefore, we are providing the original image for your inspection.



Author response image 4.

<https://doi.org/10.7554/eLife.107417.2.sa0>